

AURIX

Microcontroller Infineon Software Architecture

User's Manual

ICU Driver

Release V4.5

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Microcontrollers

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Date	Version	Change Description
2017-02-09	4.5	Updated based on review comments REV_11633
2017-01-30	4.4	Updated the date format in the version history which was not aligned correctly based on JIRA 0000054991-85 Updated based on GTM code review finding that ICU init shall only be invoked after PLL is setup. Section 6.2.1 is updated.
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2016-05-25	4.1	Modified Sections 6.2 API Functions for the following - To add field 'origin' for each API. - To modify the I/O Mode field information for each based on JIRA 0000051261-231
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2012-10-19	1.5	AI00066335. ICU support of CCU6 implemented
2012-07-30	1.4	Updated Sec 6.4 for SchM changes
2012-06-13	1.3	Review REV_003581 related changes
2012-05-14	1.2	Code Structure, Sw Description & Published info Sections updated.
2012-01-23	1.1	AI UTP AI00064638 related changes.
2011-08-10	1.0	Document Improvised
2011-03-28	0.1	Draft Version

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1 Introduction

This User Manual provides information regarding the functionality, configuration parameters and API implementation for the ICU Driver.

This User Manual is intended to help the user to get acquainted with the ICU Driver implementation for the AURIX Hardware Platform. It is the documentation that describes how to use the ICU Driver.

1.1 Scope

This document addresses the following features of the ICU Driver implementation,

- AURIX microcontroller HW peripherals used to realize the Driver
- File structure
- Configuration parameters of the Driver
- Parameters published by Driver
- API Specification
- Interrupt Handlers
- Code Snippets illustrating the common use cases.
- Limitations and Assumptions

This documentation applies to the ICU Driver Variant: Variant PB.

This document addresses the driver implementation for the following variants of AURIX.

Some of the peripheral units/feature might not be available in all the variants. In such cases, the relevant information should be discarded for the variant where the unit/feature is not applicable. [Table 1](#) gives the available units/features in the variants of AURIX for driver usage

Table 1 HW Unit Used for ICU Driver

TC Variant	Module	Module units	Channels
TC29x	TIM	6	8
TC29x	CCU6	2	3
TC29x	ERU	1	8
TC27x	TIM	4	8
TC27x	CCU6	2	3
TC27x	ERU	1	8
TC26x	TIM	3	8
TC26x	CCU6	2	3
TC26x	ERU	1	8
TC23x,TC22x,TC21x	TIM	1	8
TC23x,TC22x,TC21x	CCU6	2	3
TC23x,TC22x,TC21x	ERU	1	8

1.2 Abbreviations

The abbreviations used inside this document are explained in [Table 2](#).

Table 2 Abbreviations used in this Document

Abbreviations	Explanation
μC	Microcontroller
μs	Micro Second
API	Application Programming Interface
AUTOSAR	AUTomotive Open System Architecture
DEM	Diagnostic Error Manager
DET	Development Error Tracer
ECU	Electronic Control Unit
EcuM	ECU State Manager
GPT	General Purpose Timer
HW	Hardware
ICU	Input Capture Unit
ISR	Interrupt Service Routine
MCAL	MicroController Abstraction Layer
MCU	MicroController Unit
PWM	Pulse Width Modulator
RAM	Random Access Memory
ROM	Read Only Memory
SFR	Special Function Register
SRN	Service Request Node
GTM	Generic Timer Module
ATOM	ARU Timer Output Module
ARU	Advanced Routing Unit
SW	Software
CCU6	Capture Compare Unit

1.3 Reference

This section lists down all relevant documents for this User Manual.

Some of the artifacts listed below will not contain the version number. For those artifacts, kindly refer the latest version.

- [1] Requirements of ICU Driver,
AUTOSAR_SRS_ICU_Driver.pdf AS 4.0.3, AS 3.2.1
- [2] Specification of ICU Driver,
AUTOSAR_SWS_ICU_Driver.pdf AS 4.0.3, AS 3.2.1
- [3] User Manual MCU Driver,
Aurix_MC-ISAR_UM_MCUDriver.pdf
- [4] User Manual GTM Driver,
Aurix_MC-ISAR_UM_GTMDriver.pdf
- [5] User Manual IRQ Driver,
Aurix_MC-ISAR_UM_IRQDriver.pdf
- [6] Target TC27x User Manual: tc27xC_um_V2.2.pdf
- [7] Target TC27x User Manual: tc27xD_um_V2.2.pdf
- [8] Target TC26x User Manual: tc26xB_um_V1.3.pdf
- [9] Target TC29x User Manual: tc29xB_um_V1.2_2.pdf
- [10] Target TC23x/TC22x/TC21x User Manual: tc23x_tc22x_um_v1.1.pdf

2 ICU Driver Overview

The ICU Driver uses the GTM, CCU6 and ERU peripherals of the Aurix uC platform to realize the ICU Driver functionality. The ICU Driver is used for signal edge detection (notification), signal measurement, edge counters and time stamp capture.

2.1 Software Hardware Mapping

This section introduces the user to the HW features used for implementation. The [Figure 1](#) illustrates the peripherals and their interaction with the Drivers.

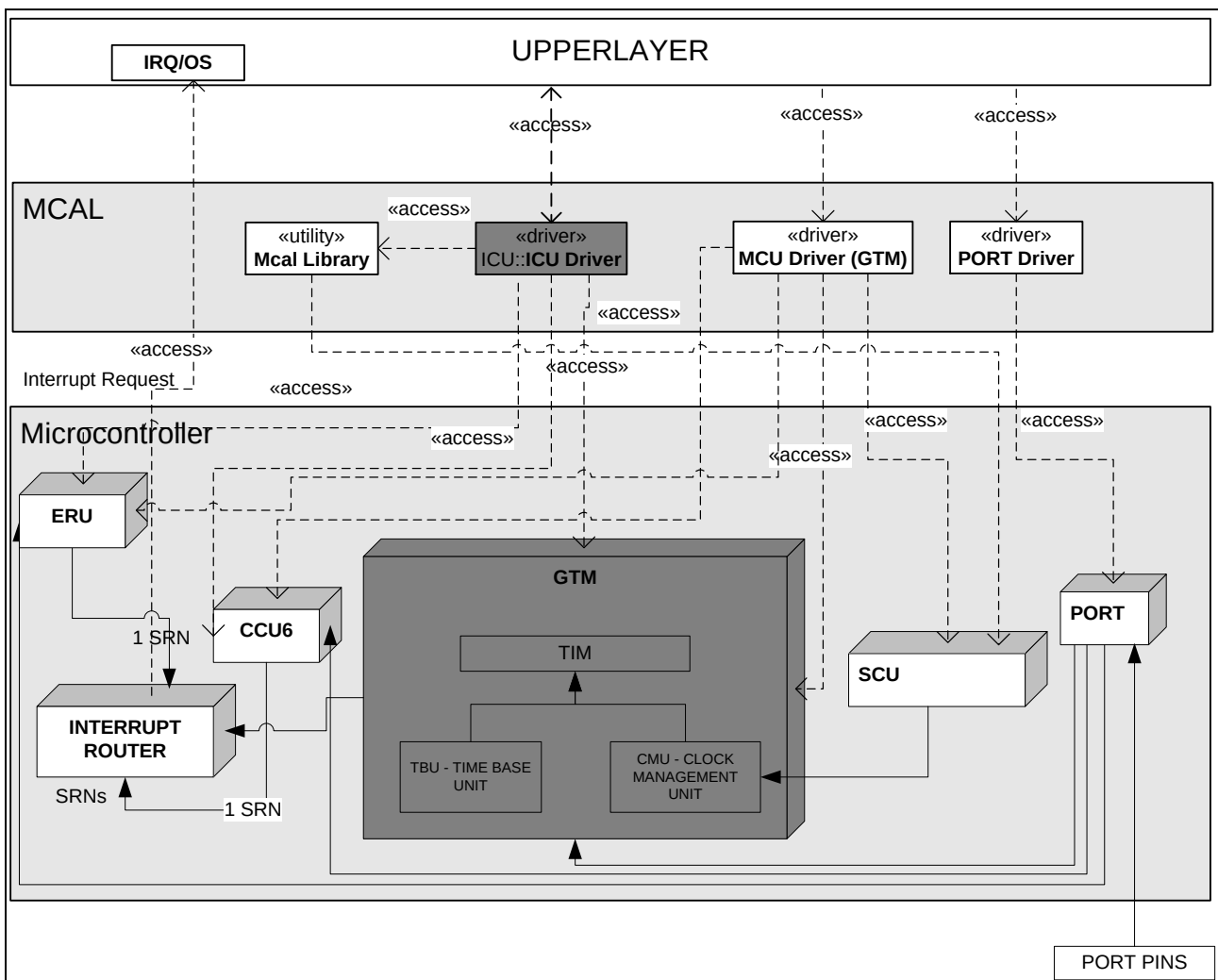


Figure 1 Software Hardware Mapping of ICU Driver.

The different HW peripherals needed by the Driver and their purpose is explained below.

The ICU Driver can access four possible resource options. The following [Table 3](#) shows the available resources for the kind of desired functionality in ICU.

Table 3 Available resources for the desired functionality in ICU

Functionality	Resources
Edge Detection (Notification)	TIM, ERU, CCU6
Signal Measurement	TIM, CCU6
Time Stamp	TIM with TBU Channel
Edge Counter	TIM

The below two TIM channel modes are used by ICU driver.

Level Mode:

- Not all the interrupts events generate a pulse on the IRQ_bit
- If IRQ_NOTIFY is set, and another interrupt arrives, the IRQ_bit stay to 1 until it is cleared (by software/hardware)

Pulse-Notify:

- Each interrupt event will also generate a pulse on the IRQ_bit signal.

Even If IRQ_NOTIFY is set, and another interrupt arrives, the IRQ_bit generates a new pulse.

The error interrupts which are part of Pulse notify mode are not supported as part of the ICU driver and to use these features GTM driver has to be used.

The [Table 4](#) shows the resource and its preferred measurement mode that could be used in ICU.

Table 4 Resources and Preferred Measurement Modes

Resource	Measurement Mode
TIM	Most recommended resource for any kind of measurement. This resource does not conflict with modules like GPT or PWM. However GTM Driver can use TIM resource separately. All four measurement modes can be implemented using TIM Module.
ERU	This resource can only be used for signal edge detection.
CCU6	This resource can only be used for Signal Edge Detection and Signal Measurement

2.1.1 ERU (External Request Unit)

The use of ERU is not limited only to the ICU driver. Additionally the ADC driver uses the common resources of ERU for gating and triggering. The ICU driver will initialize the ERU resources only used for ICU driver without affecting the other unused ERU resources.

ICU uses ERU only to detect/ notify signal edges from external signals i.e. for signal edge detection.

2.1.2 IRQ

IRQ Driver is responsible for initializing the priority of interrupts required by GTM and ERU. The mapping of TIM channels to respective SRN is given in [Table 113](#).

General Notes:

Each SRN should be provided a valid priority in the IRQ Module Configuration.

2.1.3 GTM

The GTM hardware block diagram illustrated in **Figure 2** is used as the basis for discussion of the hardware resources. The purpose of the chapter is only to focus on the resources used by the ICU Driver i.e. TIM Channels.

The use of TIM module of GTM is not limited only to the ICU Driver. Additionally GTM Complex driver use the common resources of GTM. In order to initialize these common resources of GTM, the GTM Driver is needed. GTM Driver is also responsible for clock initialization and Input/output MUX programming. The initialization of the common resources of the GTM is covered in detail in the User Manual of GTM.

To implement ICU channel, TIM Channels are used. It has to be ensured by proper configuration that same resources should not be used by GTM Complex drivers.

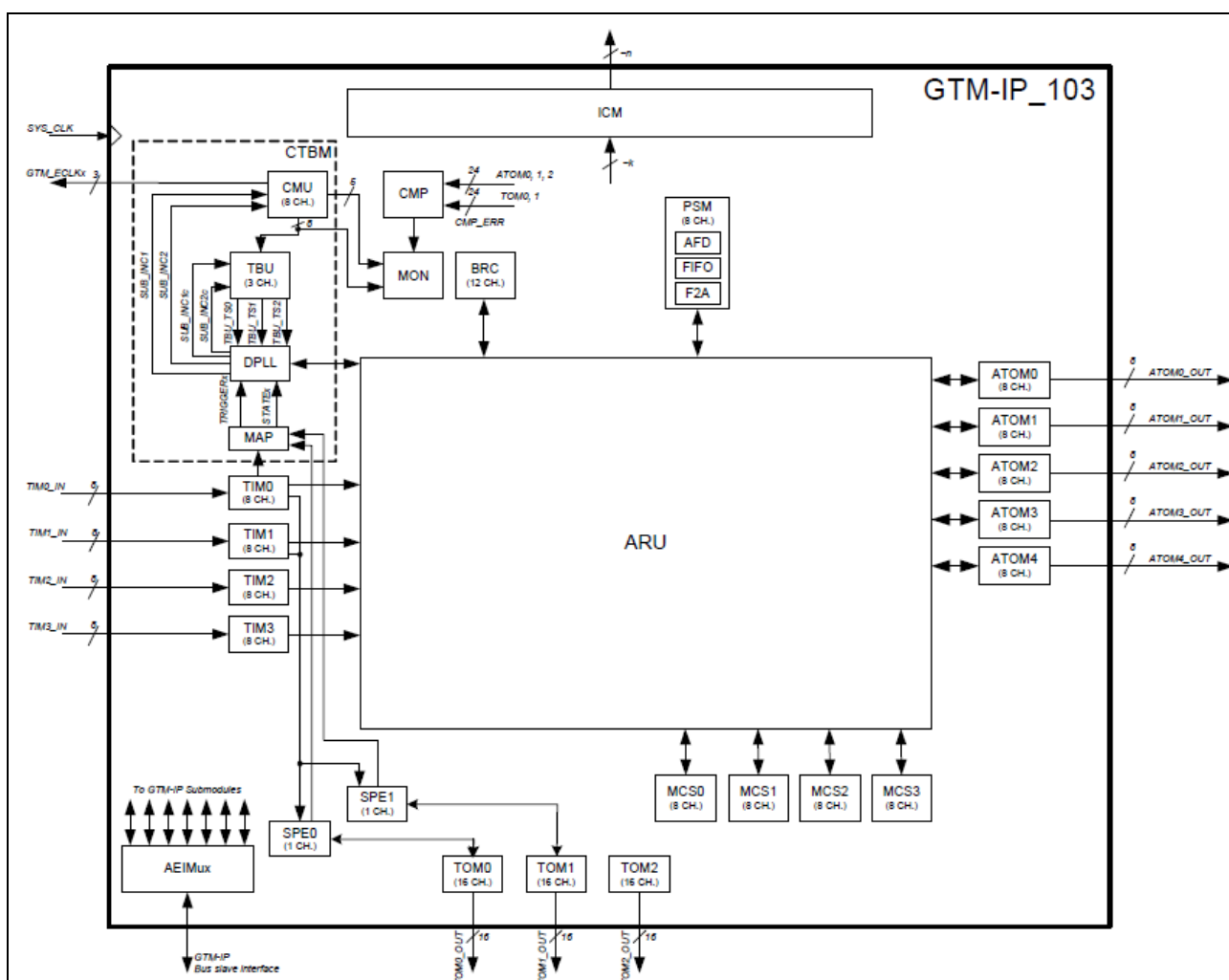


Figure 2 GTM – Hardware Block Diagram

2.1.4 SYSTEM CLOCK

The system clock is set up via the MCU Driver. It is mandatory to set up an appropriate system clock. Note that the GTM peripheral clock is derived from the system clock.

See also MCU User Manual for details about MCU configuration [3].

The clock source for timer cell is derived from the GTM clock management unit (CMU). The CMU distributes 5 fixed clock lines which clock the TOM channels and 8 Configurable Clock lines which clock the ATOM and TIM channels. The clock source configuration is part of the GTM global configuration, performed in the GTM driver configuration

See also GTM user manual for details about clock line configuration [\[4\]](#).

2.1.5 CCU6

ICU uses CCU6 only to detect/ notify signal edges from external signals i.e. for signal edge detection and measuring signal properties i.e. Signal Measurement. Capture mode of CCU6 is used to implement the functionality of ICU.

2.2 Driver Description

The [Figure 3](#) shows the Application Programming Interface (API) for the ICU Driver and its interaction with other Software Drivers.

– Development Error Tracer (DET) is required if DET is enabled for the ICU Driver.

Safety Error report will be enabled only when Safety feature is ON. As part of safety error report input range check will be done and if any error, same will be reported to SafeMcal_ReportError.

The APIs of the Driver are explained in detail in chapter [\[6\]](#)

The Configuration Data of the Driver is also depicted in the figure. The individual containers and their configuration parameters are explained in detail in chapter [\[4\]](#)



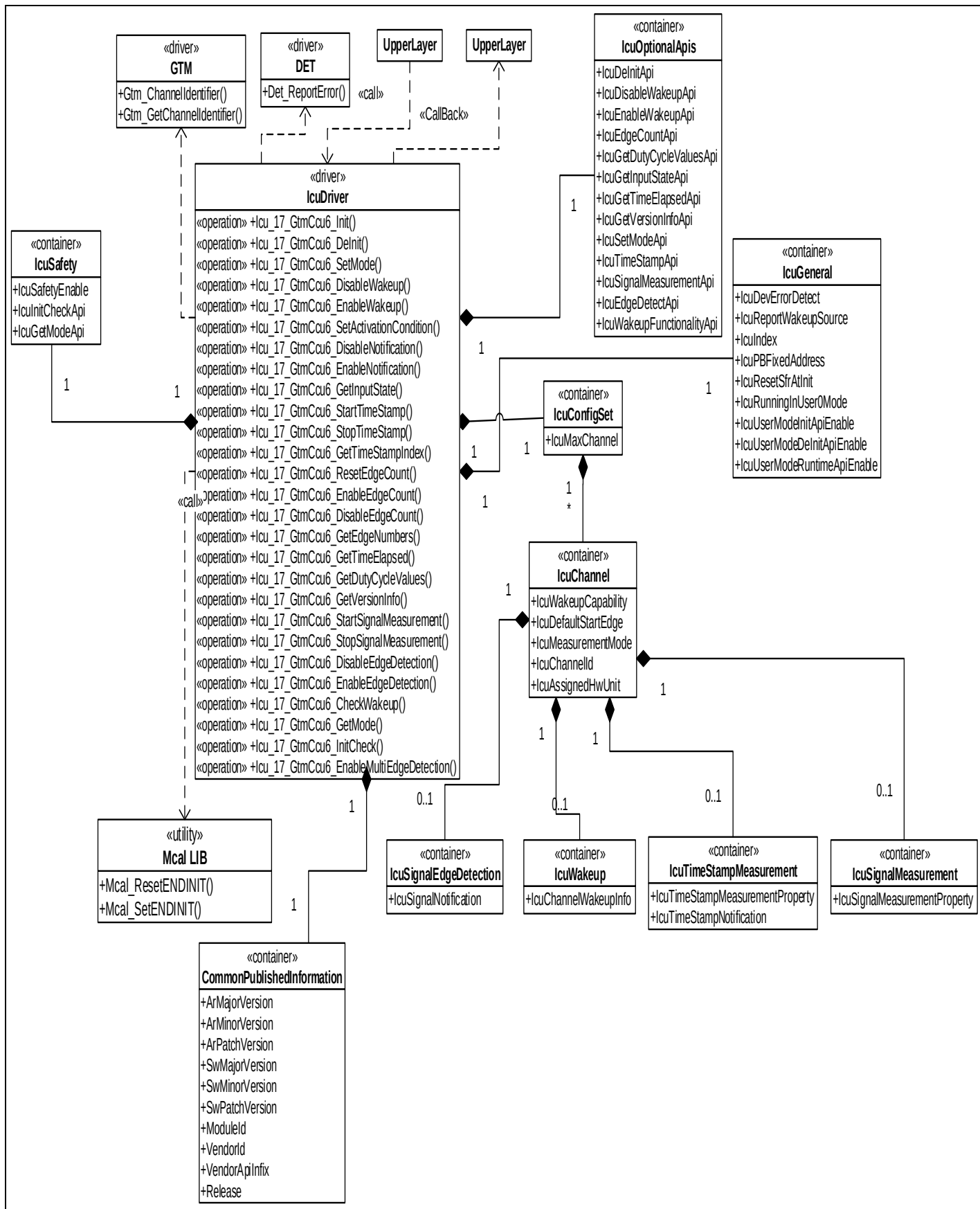
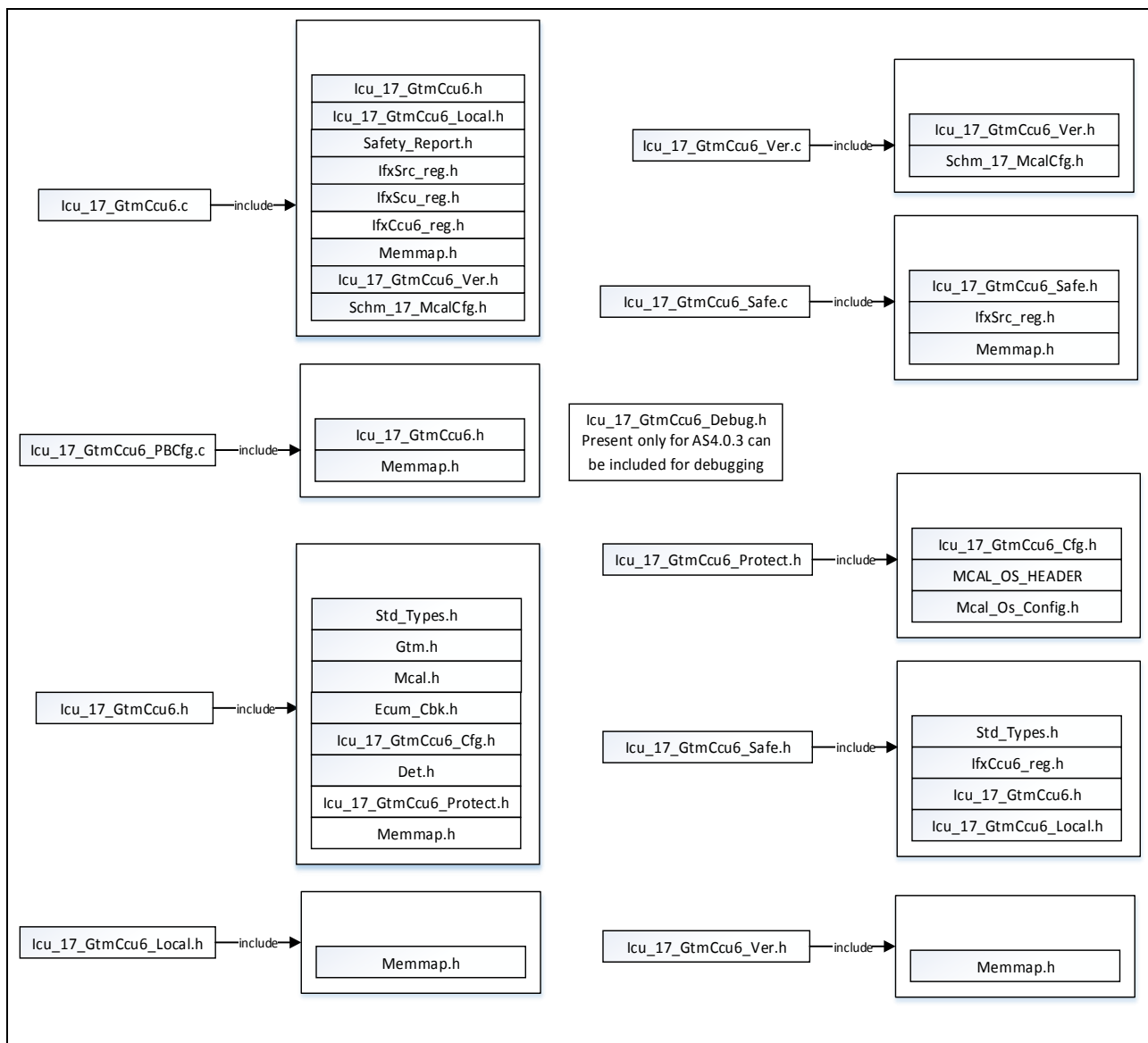


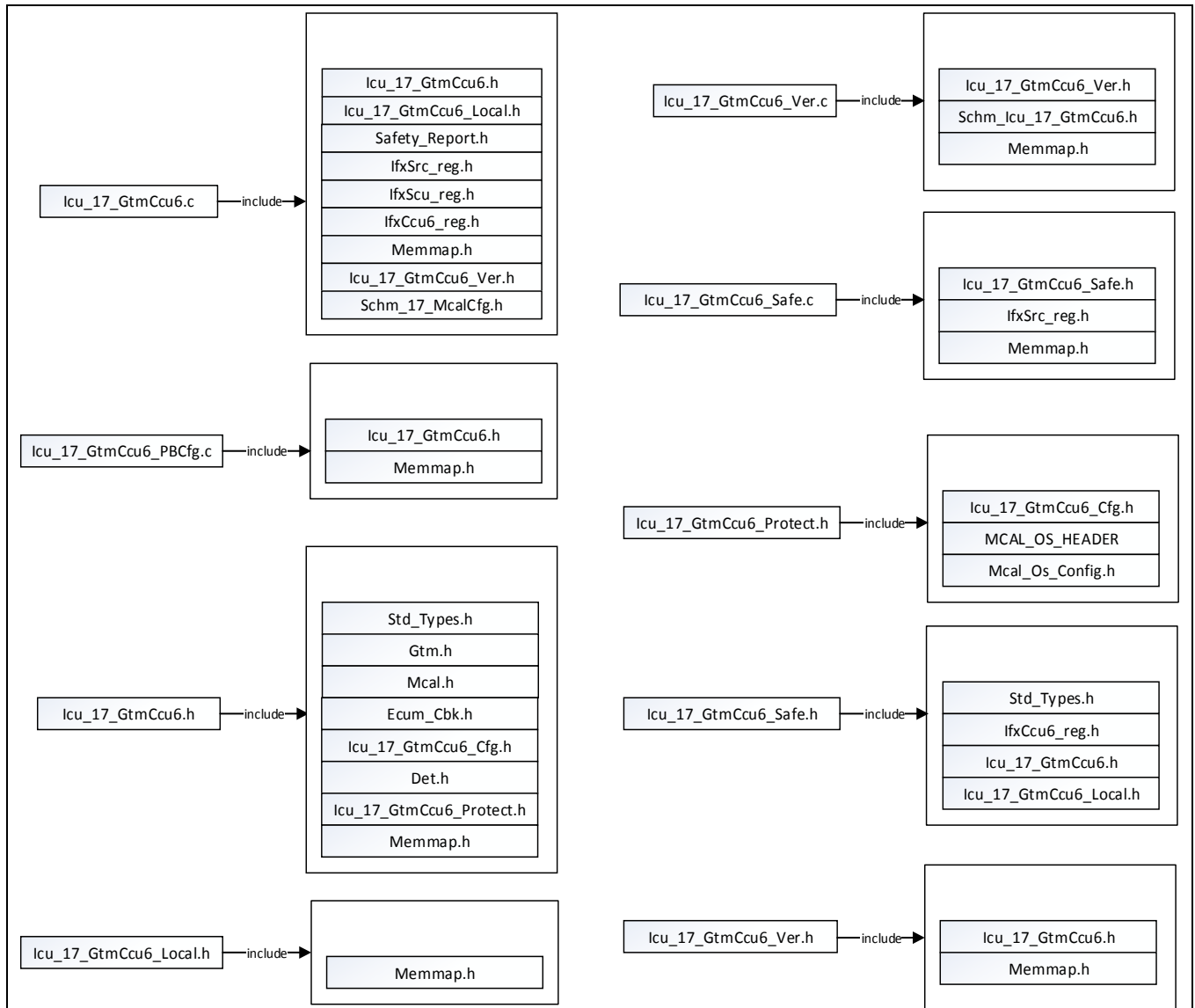
Figure 3 Interface of ICU Driver

3 File Structure

This section provides details about the ICU Driver files and other related files.

3.1 Module File Structure




Figure 4 File Structure of ICU Driver
Table 5 File Contents

File Name	File Contents
Platform_Types.h	Platform dependencies (AURIX in our case) are declared here.
Std_Types.h	Standard data types to be used are declared here.
Compiler.h	Compiler dependencies are declared here.
Compiler_Cfg.h	Contains abstraction of compiler specific keywords used for addressing data and code.
MemMap.h	Mapping of code and data (variables, constant variables) to specific memory sections
IfxSrc_reg.h	Includes the Interrupt Router register definition file for microcontroller.
IfxScu_reg.h	Includes the SCU register definition file for microcontroller.
IfxGtm_reg.h	Includes the GTM register definition file for microcontroller.
Mcal.h	Provides library functions for MCAL layer.
Det.h	Provides the functionality of Development Error Tracer

File Name	File Contents
Dem.h	Provides the functionality of Diagnostic Error Manager
Icu_17_GtmCcu6.h	This header file exports macros, type definitions and function prototypes for the ICU Driver.
Icu_17_GtmCcu6_Protect.h	Includes all the first level indirection macros for protected mode support
Icu_17_GtmCcu6.c	This file contains functionality of the Driver
Icu_17_GtmCcu6_Cfg.h	Contains precompile configuration parameters.
Icu_17_GtmCcu6_PBCfg.c	Contains post build configuration parameters.
EcuM_Cbk.h	Conditional inclusion. This file is included when a wakeup capable channel is configured to call EcuM API.
Gtm.h	This header file exports macros of GTM Driver.
SchM_Icu_17_GtmCcu6.h	This header file exports the exclusive area functions.
Icu_17_GtmCcu6_Dbg.h	Includes the extern declaration of all the Debug related variables
Icu_17_GtmCcu6_Ver.c	AUTOSAR specific functions (across 4.0.3 and 3.2.1) present here
Icu_17_GtmCcu6_Safe.c	Safe Implementation code of ICU driver exists here.

4 Configuration Documentation

This chapter summarizes

- Configuration concept used for structuring the Driver
- Configuration parameter variance w.r.t supported HW
- All configuration parameters and their description

Note: The type definition is explained for the VariantPB.

4.1 Configuration Concept

4.1.1 Configuration Class

The development of Basic Software Drivers involves the following development cycles:

- Compiling,
- Linking and
- Downloading of the executable to ECU memory.

Configuration of parameters can be done in any of these process-steps: pre-compile time, link time or even post-build time. According to the process-step that does the configuration of parameters, the configuration classes are categorized as below

Pre-compile time: The pre-compile time configuration is implemented by means of compiler switches or variables affecting the total build process. Pre-compiler configuration requires the Driver to be compiled, linked and located and hence the values of pre-compile parameters must be frozen before compilation and therefore can be used in source code distributions of the Driver only.

Link time: The Link time configuration doesn't affect compilation stage but impacts the linkage and locating stages. The Link time parameters can be used in binary (library) distributions of the Driver and the values of the Link time parameters are specified by means of external configuration source code that must be linked to the Driver during the build process.

Post-build time: The Post-build time configuration will not affect the build process and hence Post-build time configuration is applied to Driver at runtime. Since configuration location is statically independent of the Driver functionality itself but linked dynamically (e.g. at Driver initialization), post-build time configuration is used for re-programmable configuration data.

4.1.2 Configuration Variant

Based on the most suitable configuration class that may be needed for the configuration parameters of the Driver, a Driver can be delivered as the following variants

Variant PC: This variant is limited to pre-compile configuration parameters only. The intention of this variant is to optimize the parameters configuration for a source code delivery.

Variant LT: This variant allows a mix of pre-compile time- link time configuration parameters. The intention of this variant is to optimize the parameters configuration for an object code delivery.

Variant PB: This variant allows a mix of pre-compile time-, post build-time configuration parameters. The intention of this variant is to optimize the configuration parameters for a re-loadable binary.

The ICU Driver is delivered as a Variant PB.

4.1.3 How to read the Configuration Class field

The examples cited below help the user to identify the configuration class of the parameter for a given delivery type.

Note: Topics 4.1.1, 4.1.2 should be read for understanding the configuration class field

Table 6 Configuration Class Example 1

Configuration Class:	Pre-compile	x	All Variants
	Link Time	--	--
	Post build	--	--

'x' indicates that the parameter is implemented as a pre-compile parameter in all types of deliveries i.e. Variant PB, Variant LT, and Variant PC.

Table 7 Configuration Class Example 2

Configuration Class:	Pre-compile	x	Variant PC
	Link Time	--	--
	Post build	x	Variant PB

'x' indicates that the parameter is implemented as a

pre-compile parameter if the delivery type is Variant PC delivery

post-build parameter if the delivery type is a Variant PB delivery

4.2 Configuration Parameter Variance

This section describes the variance in the configuration parameter w.r.t supported microcontroller. ICU Driver doesn't have varying parameters across the supported microcontrollers.

4.3 Container: IcuGeneral

This container contains the configuration (parameters) of the ICU driver.

4.3.1 IcuDevErrorDetect

Table 8 Development Error Tracer Configuration

Name:	IcuDevErrorDetect
--------------	-------------------

Source:	This parameter is defined by the AUTOSAR, refer [2]		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Values: ON – Enabled (true) OFF – Disabled (false) Default value: OFF		
Configuration Class	Pre-Compile	x	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Switches the development error detection and notification ON or OFF. The equivalent macro (in source files) is ICU_DEV_ERROR_DETECT.		
SFRs Modified	None		
Scope:	Driver		
Dependency:	None		

4.3.2 IcuReportWakeupSource

Table 9 ICU Report WakeUp Source

Name:	IcuReportWakeupSource		
Source:	This parameter is defined by the AUTOSAR, refer [2]		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Values: ON – Report Wake up source OFF – Don't report Wakeup source Default value: OFF		
Configuration Class	Pre-Compile	x	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Switches the Wakeup Reporting functionality ON or OFF. The equivalent macro (in source files) is ICU_REPORT_WAKEUP_SOURCE. In Post Build configuration, if IcuReportWakeupSource is set and one configuration has Wakeup capable channel then all the configurations should have atleast one wakeup capable channel. If IcuReportWakeupSource is set, then GtmSleepMode should be GTM_SLEEP_DISABLE. (When Mcu sleep is requested the GTM clock will not enter into sleep mode.) In case of CCU6, CCU6 sleep module sleep on Microcontroller sleep is disabled directly by the driver itself.		
SFRs Modified	None		
Scope:	Driver		
Dependency:	None		

4.3.3 IcuPBFixedAddress

Table 10 IcuPBFixedAddress

Name:	IcuPBFixedAddress
--------------	-------------------

Source:	This parameter is microcontroller specific.		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Values: ON – PBFixedAddress feature is used OFF – PBFixedAddress feature is not used Default value: OFF		
Configuration Class	Pre-Compile	X	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Switches the PBFixedAddress functionality ON or OFF. The equivalent macro (in source files) is ICU_PB_FIXEDADDR. This feature will avoid indirections by using fixed addresses. If this feature is OFF, user can call Icu_17_GtmCcu6_Init with any of the configuration set. If this feature is ON, user can call Icu_17_GtmCcu6_Init with first configuration set.i.e Icu_17_GtmCcu6_Init(&Icu_ConfigRoot[0]);		
SFRs Modified	None		
Scope:	Driver		
Dependency:	None		

4.3.4 IcuIndex

Table 11 ICU Index Configuration

Name:	IcuIndex		
Source:	This parameter is defined by the AUTOSAR, refer [2]		
File:	-		
Range:	Values: 0 - 255 Default value: 0		
Configuration Class	Pre-Compile	X	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Specifies the InstanceId of this module instance. If only one instance is present it shall have the Id 0		
SFRs Modified	None		
Scope:	Driver		
Dependency:	None		

4.3.5 IcuDebugSupport

Table 12 IcuDebugSupport Configuration

Name:	IcuDebugSupport		
Source:	This parameter is defined by Infineon		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Values:		

	ON – Enabled (true) OFF – Disabled (false) Default value: OFF		
Configuration Class	Pre-Compile	X	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Enables debugging support for the module. This shall provide the variables that are relevant for debugging purpose as Global variables.		
SFRs Modified	None		
Scope:	Driver		
Dependency:	None		

4.3.6 IcuResetSfrAtInit

Table 13 IcuResetSfrAtInit Configuration

Name:	IcuResetSfrAtInit		
Source:	This parameter is defined by Infineon		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Values: ON – Enabled (true) OFF – Disabled (false) Default value: OFF		
Configuration Class	Pre-Compile	X	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Pre-processor switch to enable/disable resetting CCU6 and ERU SFRs to its reset value in the Icu_17_GtmCcu6_Init() API, before being used by ICU driver. The resetting of the GTM TIM channel SFRs to its reset value is done as part of the GTM driver. The Syntax of the generated parameter is, <pre>#define ICU_17_GTMCCU6_RESET_SFR_AT_INIT (STD_ON)</pre> or <pre>#define ICU_17_GTMCCU6_RESET_SFR_AT_INIT (STD_OFF)</pre>		
SFRs Modified	None		
Scope:	Driver		
Dependency:	No impact on the functionality of the ICU driver with this change.		

4.3.7 IcuUserModelInitApiEnable

Table 14 IcuUserModelInitApiEnable

Name:	IcuUserModelInitApiEnable		
Source:	This is IFX specific parameter		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Values: TRUE: FALSE:		

	Default Value: FALSE		
Configuration Class	Pre-Compile	X	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Pre-processor switch to enable / disable protected register access in all the Init APIs		
SFRs Modified	None		
Scope:	Driver		
Dependency:	None		

4.3.8 IcuUserModeDeInitApiEnable

Table 15 IcuUserModeDeInitApiEnable

Name:	IcuUserModeDeInitApiEnable		
Source:	This is IFX specific parameter		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Values: TRUE: FALSE: Default Value: FALSE		
Configuration Class	Pre-Compile	X	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Pre-processor switch to enable / disable protected register access in all the DeInit API		
SFRs Modified	None		
Scope:	Driver		
Dependency:	None		

4.3.9 IcuUserModeRuntimeApiEnable

Table 16 IcuUserModeRuntimeApiEnable

Name:	IcuUserModeRuntimeApiEnable		
Source:	This is IFX specific parameter		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Values: TRUE: FALSE: Default Value: FALSE		
Configuration Class	Pre-Compile	X	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Pre-processor switch to enable / disable protected register access in all the Runtime APIs other than Init/Deinit APIs		
SFRs Modified	None		
Scope:	Driver		
Dependency:	None		

4.3.10 IcuRunningInUser0Mode

Table 17 IcuRunningInUser0Mode

Name:	IcuRunningInUser0Mode		
Source:	This is IFX specific parameter		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Values: TRUE: FALSE: Default Value: FALSE		
Configuration Class	Pre-Compile	X	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Pre-processor switch to enable/disable the Normal/Protected mode		
SFRs Modified	None		
Scope:	Driver		
Dependency:	None		

4.4 Container: IcuSafety

4.4.1 IcuSafetyEnable

Table 18 IcuSafetyEnable

Syntax:	IcuSafetyEnable		
Source:	This parameter is defined by IFX		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Values: TRUE - Safety features enabled for ICU driver FALSE - Safety error detection disabled Default value: FALSE		
Configuration class	Pre-compile	X	All Variants
	Link Time	--	--
	Post build	--	--
Description:	Pre-processor switch for enabling the safety features of ICU driver. By enabling this flag, implicitly the following safety features are enabled: • Range check on API parameters is done by ICU driver. • ICU configuration structure will have a unique marker field which will be verified by the driver. The only dependency on marker generation is IcuSafetyEnable configuration parameter. • Safety usage of ICU driver for user to set the respective MPU settings will be published in Icu_17_GtmCcu6_ConfigDoc.html		
SFRs Modified	None		
Scope:	Driver.		
Dependency:	None		

4.4.2 IcuInitCheckApi

Table 19 IcuInitCheckApi

Syntax:	IcuInitCheckApi		
Source:	This parameter is defined by IFX		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Values: TRUE - Icu_17_GtmCcu6_InitCheck() API will be enabled FALSE - Icu_17_GtmCcu6_InitCheck() API will be disabled Default value: TRUE(Depends on IcuSafetyEnable , refer the dependency section below)		
Configuration class	Pre-compile	X	All Variants
	Link Time	--	--
	Post build	--	--
Description:	Pre-processor switch for enabling the safety feature Icu_17_GtmCcu6_InitCheck() which verifies the initialization done by ICU driver.		
SFRs Modified	None		
Scope:	Driver		
Dependency:	InitCheckApi depend on IcuSafetyEnable (refer 4.4.1) i.e If IcuSafetyEnable is enabled then only InitCheckApi is usable.		

4.4.3 IcuGetModeApi

Table 20 IcuGetModeApi

Syntax:	IcuGetModeApi		
Source:	This parameter is defined by IFX		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Values: TRUE - Icu_17_GtmCcu6_GetMode() API will be enabled FALSE - Icu_17_GtmCcu6_GetMode() API will be disabled Default value: TRUE(Depends on IcuSafetyEnable , refer the dependency section below)		
Configuration class	Pre-compile	X	All Variants
	Link Time	--	--
	Post build	--	--
Description:	Pre-processor switch for enabling the safety feature Icu_17_GtmCcu6_GetMode() which verifies the power management state of the CPU.		
SFRs Modified	None		
Scope:	Driver		
Dependency:	IcuGetModeApi depend on IcuSafetyEnable (refer 4.4.1) i.e If IcuSafetyEnable is enabled then only IcuGetModeApi is usable.		

4.5 Container: IcuOptionalApis

4.5.1 IcuDelInitApi

Table 21 ICU DelInit API

Name:	IcuDelInitApi		
Source:	This parameter is defined by the AUTOSAR, refer [2]		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Values: ON – Enabled (true) OFF – Disabled (false) Default value: OFF		
Configuration Class	Pre-Compile	x	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Switches the usage of IcuDelInitApi ON or OFF. The configuration generates this parameter as ICU_DE_INIT_API.		
SFRs Modified	Depends on the configuration set. For the current configuration the SFR's are listed in the Icu_17_GtmCcu6_ConfigDoc.html		
Scope:	Driver		
Dependency:	None		

4.5.2 IcuSetModeApi

Table 22 ICU Set Mode API

Name:	IcuSetModeApi		
Source:	This parameter is defined by the AUTOSAR, refer [2]		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Values: ON – Enabled (true) OFF – Disabled (false) Default value: OFF		
Configuration Class	Pre-Compile	x	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Switches the usage of Icu_17_GtmCcu6_SetMode API ON or OFF. The configuration generates this parameter as ICU_SET_MODE_API.		
SFRs Modified	Depends on the configuration set. For the current configuration the SFR's are listed in the Icu_17_GtmCcu6_ConfigDoc.html		
Scope:	Driver		
Dependency:	None		

4.5.3 IcuDisableWakeupApi

Table 23 ICU Disable Wakeup API

Name:	IcuDisableWakeupApi		
Source:	This parameter is defined by the AUTOSAR, refer [2]		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Values: ON – Enabled (true) OFF – Disabled (false) Default value: OFF		
Configuration Class	Pre-Compile	x	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Switches the usage of Icu_17_GtmCcu6_DisableWakeup API ON or OFF. The configuration generates this parameter as ICU_DISABLE_WAKEUP_API.		
SFRs Modified	Depends on the configuration set. For the current configuration the SFR's are listed in the Icu_17_GtmCcu6_ConfigDoc.html		
Scope:	Driver		
Dependency:	None		

4.5.4 IcuEnableWakeupApi

Table 24 Icu Enable WakeUp API

Name:	IcuEnableWakeupApi		
Source:	This parameter is defined by the AUTOSAR, refer [2]		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Values: ON – Enabled (true) OFF – Disabled (false) Should be ON if IcuWakeupCapability of any one of the channels is TRUE Should be OFF if IcuWakeupCapability is false for all ICU channels Default value: OFF		
Configuration Class	Pre-Compile	x	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Switches the usage of Icu_17_GtmCcu6_EnableWakeup API ON or OFF. The configuration generates this parameter as ICU_ENABLE_WAKEUP_API.		
SFRs Modified	Depends on the configuration set. For the current configuration the SFR's are listed in the Icu_17_GtmCcu6_ConfigDoc.html		
Scope:	Driver		
Dependency:	None		

4.5.5 IcuGetInputStateApi

Table 25 ICU Get Input State API

Name:	IcuGetInputStateApi		
Source:	This parameter is defined by the AUTOSAR, refer [2]		
File:	Icu_17_GtmCcu6_Cfg.h		

Range:	Values: ON – Enabled (true) OFF – Disabled (false) Default value: OFF		
Configuration Class	Pre-Compile	x	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Switches the usage of Icu_17_GtmCcu6_GetInputState API ON or OFF. The Configuration generates this parameter as ICU_GET_INPUT_STATE_API.		
SFRs Modified	Depends on the configuration set. For the current configuration the SFR's are listed in the Icu_17_GtmCcu6_ConfigDoc.html		
Scope:	Driver		
Dependency:	None		

4.5.6 IcuGetTimeElapsedApi

Table 26 ICU Get Time Elapsed API

Name:	IcuGetTimeElapsedApi		
Source:	This parameter is defined by the AUTOSAR, refer [2]		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Should be OFF if no channel has IcuMeasurementMode as ICU_MODE_SIGNAL_MEASUREMENT. Should be ON if at least one channel has IcuMeasurementMode as ICU_MODE_SIGNAL_MEASUREMENT and IcuSignalMeasurementProperty as ICU_PERIOD_TIME or ICU_HIGH_TIME or ICU_LOW_TIME. Values: ON – Enabled (true) OFF – Disabled (false) Default value: OFF		
Configuration Class	Pre-Compile	x	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Enabled ON – Icu_17_GtmCcu6_GetTimeElapsed () can be used. Disabled OFF – Icu_17_GtmCcu6_GetTimeElapsed () cannot be used. The configuration generates this parameter as ICU_GET_TIME_ELAPSED_API.		
SFRs Modified	Depends on the configuration set. For the current configuration the SFR's are listed in the Icu_17_GtmCcu6_ConfigDoc.html		
Scope:	Driver		
Dependency:	IcuMeasurementMode		

4.5.7 IcuGetDutyCycleValuesApi

Table 27 ICU Get Duty Cycle Values API

Name:	IcuGetDutyCycleValuesApi		
Source:	This parameter is defined by the AUTOSAR, refer [2]		
File:	Icu_17_GtmCcu6_Cfg.h		

Range:	Should be OFF if no channel has IcuMeasurementMode as ICU_MODE_SIGNAL_MEASUREMENT. Should be ON if at least one channel has IcuMeasurementMode as ICU_MODE_SIGNAL_MEASUREMENT and IcuSignalMeasurementProperty as ICU_ACTIVE_TIMEICU_DUTY_CYCLE Values: ON – Enabled (true) OFF – Disabled (false) Default value: OFF		
Configuration Class	Pre-Compile	x	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Enabled ON – Icu_17_GtmCcu6_GetDutyCycleValues () can be used. Disabled OFF – Icu_17_GtmCcu6_GetDutyCycleValues () cannot be used. The configuration generates this parameter as ICU_GET_DUTY_CYCLE_VALUES_API.		
SFRs Modified	Depends on the configuration set. For the current configuration the SFR's are listed in the Icu_17_GtmCcu6_ConfigDoc.html		
Scope:	Driver		
Dependency:	None		

4.5.8 IcuGetVersionInfoApi

Table 28 ICU Get Version Info API

Name:	IcuGetVersionInfoApi		
Source:	This parameter is defined by the AUTOSAR, refer [2]		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Values: ON – Enabled (true) OFF – Disabled (false) Default value: OFF		
Configuration Class	Pre-Compile	x	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Switches the usage of Icu_17_GtmCcu6_GetVersionInfo API ON or OFF. The Configuration generates this parameter as ICU_GET_VERSION_INFO_API.		
SFRs Modified	None		
Scope:	Driver		
Dependency:	None		

4.5.9 IcuTimestampApi

Table 29 ICU Time Stamp API

Name:	IcuTimestampApi		
Source:	This parameter is defined by the AUTOSAR, refer [2]		
File:	Icu_17_GtmCcu6_Cfg.h		

Range:	Should be OFF if no channel has IcuMeasurementMode as ICU_MODE_TIMESTAMP Should be ON if at least one channel has IcuMeasurementMode as ICU_MODE_TIMESTAMP Values: ON – Enabled (true) OFF – Disabled (false) Default value: OFF		
Configuration Class	Pre-Compile	x	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Icu_17_GtmCcu6_StartTimestamp (), Icu_17_GtmCcu6_StopTimestamp (), Icu_17_GtmCcu6_GetTimestampIndex () API's can be used if the macro is enabled. The Configuration generates this parameter as ICU_TIMESTAMP_API.		
SFRs Modified	Depends on the configuration set. For the current configuration the SFR's are listed in the Icu_17_GtmCcu6_ConfigDoc.html		
Scope:	Driver		
Dependency:	IcuMeasurementMode		

4.5.10 IcuEdgeCountApi

Table 30 ICU Edge Count API

Name:	IcuEdgeCountApi		
Source:	This parameter is defined by the AUTOSAR, refer [2]		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Should be OFF if no channel has IcuMeasurementMode as ICU_MODE_EDGE_COUNTER. Should be ON if at least one channel has IcuMeasurementMode as ICU_MODE_EDGE_COUNTER. Values: ON – Enabled (true) OFF – Disabled (false) Default value: OFF		
Configuration Class	Pre-Compile	x	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Icu_17_GtmCcu6_ResetEdgeCount (), Icu_17_GtmCcu6_EnableEdgeCount (), Icu_17_GtmCcu6_DisableEdgeCount (), Icu_17_GtmCcu6_GetEdgeNumbers () API's can be used if the macro is enabled.		
SFRs Modified	Depends on the configuration set. For the current configuration the SFR's are listed in the Icu_17_GtmCcu6_ConfigDoc.html		
Scope:	Driver		
Dependency:	IcuMeasurementMode		

4.5.11 IcuEdgeDetectionApi

Table 31 ICU Edge Detection API

Name:	IcuEdgeDetectionApi		
Source:	This parameter is defined by the AUTOSAR, refer [2]		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Should be OFF if no channel has IcuMeasurementMode as ICU_MODE_SIGNAL_EDGE_DETECT. Should be ON if at least one channel has IcuMeasurementMode as ICU_MODE_SIGNAL_EDGE_DETECT. Values: ON – Enabled (true) OFF – Disabled (false) Default value: ON		
Configuration Class	Pre-Compile	x	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	1) Optimizes the ICU Driver API's related to Edge Detection functionality. 2) Icu_17_GtmCcu6_DisableEdgeDetection () and Icu_17_GtmCcu6_EnableEdgeDetection () API's can be used if the macro is enabled. 3) Icu_17_GtmCcu6_EnableMultiEdgeDetection () API can be used if the macro is enabled.		
SFRs Modified	Depends on the configuration set. For the current configuration the SFR's are listed in the Icu_17_GtmCcu6_ConfigDoc.html		
Scope:	Driver		
Dependency:	IcuMeasurementMode		

4.5.12 IcuSignalMeasurementApi

Table 32 ICU Signal Measurement API

Name:	IcuSignalMeasurementApi		
Source:	This parameter is defined by the AUTOSAR, refer [2]		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Should be OFF if no channel has IcuMeasurementMode as ICU_MODE_SIGNAL_MEASUREMENT. Should be if ON at least one channel has IcuMeasurementMode as ICU_MODE_SIGNAL_MEASUREMENT. Values: ON – Enabled (true) OFF – Disabled (false) Default value: OFF		
Configuration Class	Pre-Compile	x	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Adds / removes the services Icu_17_GtmCcu6_StartSignalMeasurement () and Icu_17_GtmCcu6_StopSignalMeasurement () from the code.		
SFRs Modified	Depends on the configuration set. For the current configuration the SFR's are listed in the Icu_17_GtmCcu6_ConfigDoc.html		

Scope:	Driver
Dependency:	IcuMeasurementMode

4.5.13 IcuWakeupFunctionalityApi

Table 33 ICU Wakeup Functionality API

Name:	IcuWakeupFunctionalityApi		
Source:	This parameter is defined by the AUTOSAR, refer [2]		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Values: ON – Enabled (true) OFF – Disabled (false) Default value: OFF		
Configuration Class	Pre-Compile	x	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Adds / removes the service Icu_17_GtmCcu6_CheckWakeup () from the code.		
SFRs Modified	Depends on the configuration set. For the current configuration the SFR's are listed in the Icu_17_GtmCcu6_ConfigDoc.html		
Scope:	Driver		
Dependency:	None		

4.6 Container: IcuConfigSet

This container contains Channel configurations. IcuConfigSet will have atleast one Configuration set and user can configure number of configsets. Each IcuConfigSet should have a corresponding MCU configuration set, i.e. If user configures 3 IcuConfigSet's ,he should have 3 MCU configurations and it is one to one mapped.

4.6.1 IcuMaxChannel

Table 34 IcuMaxChannel

Name:	IcuMaxChannel		
Source:	This parameter is defined by the AUTOSAR, refer [2]		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Values: 0 to Maximum number of ICU channels possible. Default value: 0		
Configuration Class	Pre-Compile	--	--
	Link Time	--	--
	Post Build	x	Variant PB
Description:	Maximum number of channels configured in the current channel configuration set. This value is derived from the channel configuration sets configured by the user. User need to provide the number of configured channels in the current Config Set. This information is not used by the driver. ICU_MAX_CHANNELS #define is generated by the driver which is derived from all the Channel Configuration sets configured by the user.		
SFRs Modified	None		

Scope:	Driver
Dependency:	None

4.6.2 Container: IcuChannel

4.6.2.1 IcuChannelId

Name:	IcuChannelId		
Source:	This parameter is defined by the AUTOSAR, refer [2]		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Unique number to identify the channel in the channel configuration set. Values: Any Values between 0 to maximum number of channels in same channel configuration set. Default value: Index value in the IcuChannel container. i.e. in sequential order.		
Configuration Class	Pre-Compile	--	--
	Link Time	--	--
	Post Build	x	Variant PB
Description:	Channel Id of the ICU channel. This value will be assigned to the symbolic name derived of the IcuChannel container short name. If the Symbolic name assigned to the channel is "MAIN_BRAKE", then the symbolic name will be available in the format "ICU_MAIN_BRAKE"		
SFRs Modified	None		
Scope:	Driver		
Dependency:	None		

4.6.2.2 IcuWakeupCapability

Table 35 ICU Wakeup Capability

Name:	IcuWakeupCapability		
Source:	This parameter is defined by the AUTOSAR, refer [2]		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Significant if IcuMeasurementMode is IcuSignalEdgeDetection Values: TRUE FALSE Default value: FALSE		
Configuration Class	Pre-Compile	--	--
	Link Time	--	--
	Post Build	x	Variant PB
Description:	Information about the wakeup-capability of this channel.		
SFRs Modified	None		
Scope:	Driver		
Dependency:	IcuMeasurementMode		

4.6.2.3 IcuDefaultStartEdge

Table 36 ICU Default Start Edge

Name:	IcuDefaultStartEdge		
Source:	This parameter is defined by the AUTOSAR, refer [2]		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	IcuBothEdges is significant if IcuSignalMeasurementProperty is not ICU_PERIOD_TIME or ICU_ACTIVE_TIMEICU_DUTY_CYCLE Values: ICU_FALLING_EDGE, ICU_RISING_EDGE, ICU_BOTH_EDGES Default value: ICU_RISING_EDGE		
Configuration Class	Pre-Compile	--	--
	Link Time	--	--
	Post Build	x	Variant PB
Description:	This is the default edge configured to carry out the desired measurement. For Signal measurement Property, the information given here is constant and applicable throughout program. For other measurements, Icu_17_GtmCcu6_SetActivationCondition can be used to change the edges.		
SFRs Modified	None		
Scope:	Driver		
Dependency:	IcuSignalMeasurementProperty		

4.6.2.4 IcuMeasurementMode

Table 37 ICU Measurement Mode

Name:	IcuMeasurementMode		
Source:	This parameter is defined by the AUTOSAR, refer [2]		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Values: ICU_MODE_SIGNAL_EDGE_DETECT, ICU_MODE_EDGE_COUNTER, ICU_MODE_SIGNAL_MEASUREMENT, ICU_MODE_TIMESTAMP Default value: ICU_MODE_SIGNAL_EDGE_DETECT		
Configuration Class	Pre-Compile	--	--
	Link Time	--	--
	Post Build	x	Variant PB
Description:	Configuration parameter to determine the kind of measurement required for the channel.		
SFRs Modified	Depends on the configuration set. For the current configuration the SFR's are listed in the Icu_17_GtmCcu6_ConfigDoc.html		
Scope:	Driver		
Dependency:	In case of ICU_MODE_TIMESTAMP mode user should configure TBU Channel in GTM configuration (in case GTM TIM channel is used), which is used as time base for time stamp measurement. Refer 2		

4.6.2.5 IcuAssignedHwUnit

Table 38 ICU Assigned HW Unit

Name:	IcuAssignedHwUnit		
Source:	This parameter is microcontroller specific.		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Values : This is a reference parameter to a) TIMx_Channely (where x and y depends on the microcontroller) b) ERUx (where x depends on the microcontroller): Significant if IcuMeasurmentMode is ICU_MODE_SIGNAL_EDGE_DETECT c) CCU6x Channely (where x and y depends on the microcontroller): Significant if IcuMeasurementMode is ICU_MODE_SIGNAL_EDGE_DETECT or ICU_MODE_SIGNAL_MEASUREMENT Default value: None The channel chosen in IcuAssignedHwUnit should not be used by other channels of ICU or other BSW modules		
Configuration Class	Pre-Compile	--	--
	Link Time	--	--
	Post Build	x	Variant PB
Description:	The HW resource required for the measurement. TIM0: Channel 0 to Channel 7 TIM1: Channel 0 to Channel 7 TIM2: Channel 0 to Channel 7 TIM3: Channel 0 to Channel 7 ERU0 ERU1 ERU2 ERU3 ERU4 ERU5 ERU6 ERU7 CCU60: CC60 to CC62 CCU61: CC60 to CC62 Please note: If ERU is chosen, OGU0 of ERU resource will be used. Hence it must be ensured that no other module uses OGU0.		
SFRs Modified	Depends on the configuration set. For the current configuration the SFR's are listed in the Icu_17_GtmCcu6_ConfigDoc.html		
Scope:	Driver		
Dependency:	IcuMeasurementMode		

4.6.2.6 Container: IcuSignalEdgeDetection

This container contains the configuration (parameters) in case the measurement mode is "ICU_MODE_SIGNAL_EDGE_DETECT".

4.6.2.6.1 IcuSignalNotification

Table 39 ICU Signal Notification

Name:	IcuSignalNotification		
Source:	This parameter is defined by the AUTOSAR, refer [2]		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Significant only if IcuMeasurementMode is ICU_MODE_SIGNAL_EDGE_DETECT Value: User defined notification function name of type FUNCTION_NAME(Selectable) / Address (Loadable) Default value: None. If no notification is desired, then the parameter must be removed i.e. multiplicity of 0.		
Configuration Class	Pre-Compile	--	--
	Link Time	--	--
	Post Build	x	Variant PB
Description:	Notification Pointer function or callback function: When edge event occurs this function will be called by the ICU driver for each individual channel.		
SFRs Modified	Depends on the configuration set. For the current configuration the SFR's are listed in the Icu_17_GtmCcu6_ConfigDoc.html		
Scope:	Driver		
Dependency:	IcuMeasurementMode		

4.6.2.7 Container: IcuWakeup

This container contains the configuration (parameters) needed to configure a wakeup capable channel. The measurement mode should be ICU_MODE_SIGNAL_EDGE_DETECT.

4.6.2.7.1 IcuChannelWakeupInfo

Table 40 ICU Channel Wakeup Info

Name:	IcuChannelWakeupInfo		
Source:	This parameter is defined by the AUTOSAR, refer [2]		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Significant only if IcuWakeupCapability is true and IcuMeasurementMode is ICU_MODE_SIGNAL_EDGE_DETECT Values : This is a reference parameter of type EcuMWakeupSource Default value: None		
Configuration Class	Pre-Compile	--	--
	Link Time	--	--
	Post Build	X	Variant PB
Description:	When ICU in wakeup mode and edge occurs on wakeup capable channel, function EcuM_CheckWakeup () will be called with this parameter.		
SFRs Modified	None		
Scope:	Driver		
Dependency:	IcuWakeupCapability, IcuMeasurementMode		

4.6.2.8 Container: IcuSignalMeasurement

This container contains the configuration in case the measurement mode is "ICU_MODE_SIGNAL_MEASUREMENT"

4.6.2.8.1 IcuSignalMeasurementProperty

Table 41 ICU Signal Measurement Property

Name:	IcuSignalMeasurementProperty		
Source:	This parameter is defined by the AUTOSAR, refer [2]		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Significant only if IcuMeasurementMode is ICU_MODE_SIGNAL_MEASUREMENT Values: <START AS40>ICU_ACTIVE_TIME<END AS40><START AS321>ICU_DUTY_CYCLE<END AS321>, ICU_HIGH_TIME, ICU_LOW_TIME, ICU_PERIOD_TIME Default value: None		
Configuration Class	Pre-Compile	--	--
	Link Time	--	--
	Post Build	X	Variant PB
Description:	Configuration parameter to determine the kind of signal measurement required for the channel. <START AS40>ICU_DUTY_CYCLE is not provided according to ongoing Bugzilla 52843.<END AS40>		
SFRs Modified	Depends on the configuration set. For the current configuration the SFR's are listed in the Icu_17_GtmCcu6_ConfigDoc.html		
Scope:	Driver		
Dependency:	IcuMeasurementMode		

4.6.2.8.2 IcuDutycycleBufferMarker

Table 42 Specification of IcuDutycycleBufferMarker

Name:	IcuDutycycleBufferMarker		
Source:	This parameter is defined by IFX		
File:	Icu_17_GtmCcu6_PBCfg.c		
Range:	0 – 65535		
Default value:	0		
Configuration Class:	Pre-Compile	--	--
	Link Time	--	--
	Post Build	X	Variant PB
Description:	The parameter defines the marker value of the Buffer per Signal measurement channel. The marker is used for validating the correct buffer pointer. This parameter is significant only if IcuSafetyEnable is set to True. For a QM channel this parameter is non-editable and will have the default value 0. Even if a non-zero value is configured, during configuration generation this value will be discarded and 0 will be		

	generated.
SFRs Modified	None
Scope:	Driver
Dependency:	IcuSafetyEnable

4.6.2.9 Container: IcuTimestampMeasurement

This container contains the configuration (parameters) in case the measurement mode is "ICU_MODE_TIMESTAMP"

4.6.2.9.1 IcuTimestampMeasurementProperty

Table 43 ICU Time Stamp Measurement Property

Name:	IcuTimestampMeasurementProperty		
Source:	This parameter is defined by the AUTOSAR, refer [2]		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Significant only if IcuMeasurementmode is ICU_MODE_TIMESTAMP Values: ICU_LINEAR_BUFFER, ICU_CIRCULAR_BUFFER Default value: None		
Configuration Class	Pre-Compile	--	--
	Link Time	--	--
	Post Build	x	Variant PB
Description:	Configuration parameter to determine the kind of timestamp measurement required for the channel.		
SFRs Modified	TIMi_CHx_CTRL, TIMi_CHx_IRQ_NOTIFY, TIMi_CHx_IRQ_EN		
Scope:	Driver		
Dependency:	IcuMeasurementmode		

4.6.2.9.2 IcuTimestampNotification

Table 44 ICU Time Stamp Notification

Name:	IcuTimeStampNotification		
Source:	This parameter is defined by the AUTOSAR, refer [2]		
File:	Icu_17_GtmCcu6_Cfg.h		
Range:	Significant only if IcuMeasurementmode is ICU_MODE_TIMESTAMP Value: User defined notification function name of type FUNCTION_NAME(Selectable) / Address (Loadable) Default value: NULL		
Configuration Class	Pre-Compile	--	--
	Link Time	--	--
	Post Build	x	Variant PB
Description:	Notification pointer function or callback function. The same variable is also used for signal edge notification. One channel cannot be in two modes at the same time.		
SFRs Modified	None		

Scope:	Driver
Dependency:	IcuMeasurementmode

4.6.2.9.3 IcuTimestampBufferMarker

Table 45 Specification of IcuTimestampBufferMarker

Name:	IcuTimestampBufferMarker		
Source:	This parameter is defined by IFX		
File:	Icu_17_GtmCcu6_PBCfg.c		
Range:	0 – 65535		
Default value:	0		
Configuration Class:	Pre-Compile	--	--
	Link Time	--	--
	Post Build	x	Variant PB
Description:	The parameter defines the marker value of the Buffer per timestamp measurement channel. The marker is used for validating the correct buffer pointer. This parameter is significant only if IcuSafetyEnable is set to True. Also, for a QM channel this parameter is non-editable and will have the default value 0. Even if a non-zero value is configured, during configuration generation this value will be discarded and 0 will be generated		
SFRs Modified	None		
Scope:	Driver		
Dependency:	IcuSafetyEnable		

4.7 Container: CommonPublishedInformation

It contains published information about vendor and versions.

4.7.1 ArMajorVersion

Table 46 ArMajorVersion

Name:	ArMajorVersion		
Source:	This is Infineon specific parameter.		
File:	Icu_17_GtmCcu6.xdm/ Icu_17_GtmCcu6.bmd		
Range:	-		
Configuration Class	Pre-Compile	x	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Major version number of AUTOSAR specification on which the appropriate implementation is based on.		
SFRs Modified	None		
Scope:	Driver		
Dependency:	None		

4.7.2 ArMinorVersion

Table 47 ArMinorVersion

Name:	ArMinorVersion		
Source:	This is Infineon specific parameter.		
File:	Icu_17_GtmCcu6.xdm/ Icu_17_GtmCcu6.bmd		
Range:	-		
Configuration Class	Pre-Compile	x	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Minor version number of AUTOSAR specification on which the appropriate implementation is based on.		
SFRs Modified	None		
Scope:	Driver		
Dependency:	None		

4.7.3 ArPatchVersion

Table 48 ArPatchVersion

Name:	ArPatchVersion		
Source:	This is Infineon specific parameter.		
File:	Icu_17_GtmCcu6.xdm/Icu_17_GtmCcu6.bmd		
Range:	-		
Configuration Class	Pre-Compile	x	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Patch level version number of AUTOSAR specification on which the appropriate implementation is based on.		
SFRs Modified	None		
Scope:	Driver		
Dependency:	None		

4.7.4 SwMajorVersion

Table 49 SwMajorVersion

Name:	SwMajorVersion		
Source:	This is Infineon specific parameter.		
File:	Icu_17_GtmCcu6.xdm/Icu_17_GtmCcu6.bmd		
Range:	-		
Configuration Class	Pre-Compile	x	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Major version number of the vendor specific implementation of the module.		
SFRs Modified	None		
Scope:	Driver		

Dependency:	None
--------------------	------

4.7.5 SwMinorVersion

Table 50 SwMinorVersion

Name:	SwMinorVersion		
Source:	This is Infineon specific parameter.		
File:	Icu_17_GtmCcu6.xdm/Icu_17_GtmCcu6.bmd		
Range:	-		
Configuration Class	Pre-Compile	x	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Minor version number of the vendor specific implementation of the module.		
SFRs Modified	None		
Scope:	Driver		
Dependency:	None		

4.7.6 SwPatchVersion

Table 51 SwPatchVersion

Name:	SwMajorVersion		
Source:	This is Infineon specific parameter.		
File:	Icu_17_GtmCcu6.xdm/Icu_17_GtmCcu6.bmd		
Range:	-		
Configuration Class	Pre-Compile	x	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Patch level version number of the vendor specific implementation of the module.		
SFRs Modified	None		
Scope:	Driver		
Dependency:	None		

4.7.7 ModuleId

Table 52 ModuleId

Name:	ModuleId		
Source:	This is Infineon specific parameter.		
File:	Icu_17_GtmCcu6.xdm/Icu_17_GtmCcu6.bmd		
Range:	122		
Configuration Class	Pre-Compile	x	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Module ID of ICU module from Module List		
SFRs Modified	None		

Scope:	Driver
Dependency:	None

4.7.8 VendorId

Table 53 VendorId

Name:	VendorId		
Source:	This is Infineon specific parameter.		
File:	Icu_17_GtmCcu6.xdm/Icu_17_GtmCcu6.bmd		
Range:	17		
Configuration Class	Pre-Compile	x	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	Vendor ID of the ICU module according to the AUTOSAR vendor list.		
SFRs Modified	None		
Scope:	Driver		
Dependency:	None		

4.7.9 VendorApilInfix

Table 54 VendorApilInfix

Name:	VendorApilInfix		
Source:	This is Infineon specific parameter.		
File:	Icu_17_GtmCcu6.xdm/Icu_17_GtmCcu6.bmd		
Range:	GtmCcu6		
Configuration Class	Pre-Compile	x	All Variants
	Link Time	--	--
	Post Build	--	--
Description:	In driver modules which can be instantiated several times on a single ECU then the name of APIs is extended by the VendorId and a vendor specific name. This parameter is used to specify the vendor specific name.		
SFRs Modified	None		
Scope:	Driver		
Dependency:	None		

4.7.10 Release

Table 55 Release

Name:	Release		
Source:	This is Infineon specific parameter.		
File:	Icu_17_GtmCcu6.xdm/Icu_17_GtmCcu6.bmd		
Range:	_TRICORE_TC275 _TRICORE_TC277		
Configuration	Pre-Compile	x	All Variants

Class	Link Time	--	--
	Post Build	--	--
Description:	Aurix derivative used for the implementation.		
SFRs Modified	None		
Scope:	Driver		
Dependency:	None		

5 Published Parameters

This section describes the parameters published by the ICU Driver.

5.1 VendorId

Table 56 Vendor ID

Name:	ICU_17_GTMCCU6_VENDOR_ID
Type:	Macro (value typecast to uint16)
File:	Icu_17_GtmCcu6.h
Value:	17
Description:	Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list

5.2 ModuleId

Table 57 Module ID

Name:	ICU_17_GTMCCU6_MODULE_ID
Type:	Macro (value typecast to uint16)
File:	Icu_17_GtmCcu6.h
Value:	122
Description:	Module Id

5.3 InstanceId

Table 58 Instance ID

Name:	ICU_17_GTMCCU6_INSTANCE_ID
Type:	Macro (value typecast to uint8)
File:	Icu_17_GtmCcu6_Cfg.h
Value:	Derived from IcuIndex
Description:	Instance Id of the driver

5.4 ArMajorVersion

Table 59 Software AUTOSAR Major Version

Name:	ICU_17_GTMCCU6_AR_<START AS40>RELEASE_<END AS40>MAJOR_VERSION
Type:	Macro

File:	Icu_17_GtmCcu6_Cfg.h
Value:	Refer Icu_17_GtmCcu6_Cfg.h file
Description:	Major version of AUTOSAR specification on which the appropriate implementation is based on.

5.5 ArMinorVersion

Table 60 Software AUTOSAR Minor Version

Name:	ICU_17_GTMCCU6_AR_<START AS40>RELEASE_<END AS40>MINOR_VERSION
Type:	Macro
File:	Icu_17_GtmCcu6_Cfg.h
Value:	Refer Icu_17_GtmCcu6_Cfg.h file
Description:	Minor version of AUTOSAR specification on which the appropriate implementation is based on.

5.6 ArPatchVersion

Table 61 Software AUTOSAR Patch Version

Name:	ICU_17_GTMCCU6_AR_<START AS40>RELEASE_REVISION_<END AS40><START AS321>PATCH_<END AS321>VERSION
Type:	Macro
File:	Icu_17_GtmCcu6_Cfg.h
Value:	Refer Icu_17_GtmCcu6_Cfg.h file
Description:	Patch level version of AUTOSAR specification on which the appropriate implementation is based on.

5.7 SwMajorVersion

Table 62 Software Major Version

Name:	ICU_17_GTMCCU6_SW_MAJOR_VERSION
Type:	Macro / uint8
File:	Icu_17_GtmCcu6_Cfg.h
Value:	Refer Icu_17_GtmCcu6_Cfg.h file
Description:	Major version number of the vendor specific implementation of the module.

5.8 SwMinorVersion

Table 63 Software Minor Version

Name:	ICU_17_GTMCCU6_SW_MINOR_VERSION
Type:	Macro / uint8
File:	Icu_17_GtmCcu6_Cfg.h
Value:	Refer Icu_17_GtmCcu6_Cfg.h file
Description:	Minor version number of the vendor specific implementation of the module.

5.9 SwPatchVersion

Table 64 Software Patch Version

Name:	ICU_17_GTMCCU6_SW_PATCH_VERSION
Type:	Macro / uint8
File:	Icu_17_GtmCcu6_Cfg.h
Value:	Refer Icu_17_GtmCcu6_Cfg.h file
Description:	Patch version number of the vendor specific implementation of the module.

6 API Documentation

The API documentation describes the syntax and semantics of the API functions as well as it gives some additional implementation specific behavior description not defined by the AUTOSAR ICU specification.

Implementation specific type definitions, which are required for configuration and convenience, are also documented.

6.1 API Type Definitions

This section describes all the type definitions that are exported by ICU Driver. These type definitions are either used for configuration or for API parameter passing.

6.1.1 Type Definition Icu_17_GtmCcu6_ModeType

Table 65 Type Definition Icu_17_GtmCcu6_ModeType

Syntax:	Icu_17_GtmCcu6_ModeType
Type:	uint8
File:	Icu_17_GtmCcu6.h
Range:	ICU_MODE_NORMAL (0): Normal operation ICU_MODE_SLEEP (1): Operation for reduced power mode.
Description:	Enabling / Disabling the interrupts which are not required for the ECU wakeup.

6.1.2 Type Definition Icu_17_GtmCcu6_InputStateType

Table 66 Type Definition Icu_17_GtmCcu6_InputStateType

Syntax:	Icu_17_GtmCcu6_InputStateType	
Type:	uint8	
File:	Icu_17_GtmCcu6.h	
Range:	ICU_IDLE	No activation edge has been detected since the last call of Icu_17_GtmCcu6_GetInputState () or Icu_17_GtmCcu6_Init () for signal edge detection channels. No new data has been detected since the last call of Icu_17_GtmCcu6_Init () or Icu_17_GtmCcu6_GetTimeElapsed () or

		Icu_17_GtmCcu6_GetDutyCycleValues () for signal measurement channels.
	ICU_ACTIVE	An activation edge has been detected for edge detection channels. New data is captured for signal measurement channels.
Description:	This type defines the status information from ICU driver when Icu_17_GtmCcu6_GetInputState () called.	

6.1.3 Type Definition Icu_17_GtmCcu6_ValueType

Table 67 Type Definition Icu_17_GtmCcu6_ValueType

Syntax:	Icu_17_GtmCcu6_ValueType
Type:	uint32
File:	Icu_17_GtmCcu6.h
Range:	0 to 0xFFFFFFFF
Description:	The width of the widest timer register used in the module. The TBU is 24 bits wide. The Icu_17_GtmCcu6_ValueType corresponds to width of the buffer for timestamp ticks and measured elapsed time ticks.

6.1.4 Type Definition Icu_17_GtmCcu6_DutyCycleType

Table 68 Type Definition Icu_17_GtmCcu6_DutyCycleType

Syntax:	Icu_17_GtmCcu6_DutyCycleType	
Type:	Structure	
File:	Icu_17_GtmCcu6.h	
Range:	ActiveTime (Icu_17_GtmCcu6_ValueType)	Coherent active-time measured on the channel.
	PeriodTime (Icu_17_GtmCcu6_ValueType)	Coherent period time measured on the channel.
	BufferMarker (Icu_17_GtmCcu6_ValueType)	Buffer marker value for signal measurement
Description:	A structure to fill in the duty cycle in API Icu_17_GtmCcu6_GetDutyCycleValues	

6.1.5 Type Definition Icu_17_GtmCcu6_IndexType

Table 69 Type Definition Icu_17_GtmCcu6_IndexType

Syntax:	Icu_17_GtmCcu6_IndexType
Type:	uint16
File:	Icu_17_GtmCcu6.h
Range:	0 to 0xFFFF.
Description:	Return value used in Icu_17_GtmCcu6_GetTimeStampIndex

6.1.6 Type Definition Icu_17_GtmCcu6_EdgeNumberType

Table 70 Type Definition Icu_17_GtmCcu6_EdgeNumberType

Syntax:	Icu_17_GtmCcu6_EdgeNumberType
Type:	uint32
File:	Icu_17_GtmCcu6.h
Range:	0 to 0xFFFFFFFF.
Description:	Return value used in Icu_17_GtmCcu6_GetEdgeNumbers

6.1.7 Type Definition Icu_17_GtmCcu6_ChannelType

Table 71 Type Definition Icu_17_GtmCcu6_ChannelType

Syntax:	Icu_17_GtmCcu6_ChannelType
Type:	uint8
File:	Icu_17_GtmCcu6.h
Range:	Channels 0 to 255(or the maximum number of TIM and ERU channels)
Description:	Specifies the identification (ID) for a Channel.

6.1.8 Type Definition Icu_17_GtmCcu6_Activationtype

Table 72 Type Definition Icu_17_GtmCcu6_Activationtype

Syntax:	Icu_17_GtmCcu6_Activationtype
Type:	Struct
File:	Icu_17_GtmCcu6.h
Range:	Range: ICU_RISING_EDGE: An appropriate action shall be executed when a rising edge occurs on the ICU input signal. ICU_FALLING_EDGE: An appropriate action shall be executed when a falling edge occurs on the ICU input signal. ICU_BOTH_EDGES: An appropriate action shall be executed when either a rising or falling edge occur on the ICU input signal.
Description:	Definition of the type of activation of an ICU channel.

6.1.9 Type Definition Icu_17_GtmCcu6_MeasurementModeType

Table 73 Type Definition Icu_17_GtmCcu6_MeasurementModeType

Syntax:	Icu_17_GtmCcu6_MeasurementModeType
Type:	Struct
File:	Icu_17_GtmCcu6.h
Range:	Range : ICU_MODE_SIGNAL_EDGE_DETECT : Mode for detecting edges ICU_MODE_SIGNAL_MEASUREMENT : Mode for measuring different times between various configurable edges ICU_MODE_TIMESTAMP : Mode for capturing timer values on configurable edges ICU_MODE_EDGE_COUNTER : Mode for counting edges on configurable edges

Description:	Definition of the measurement mode type
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6.1.10 Type Definition Icu_17_GtmCcu6_SignalMeasurementPropertyType

Table 74 Type Definition Icu_17_GtmCcu6_SignalMeasurementPropertyType

Syntax:	Icu_17_GtmCcu6_SignalMeasurementPropertyType
Type:	Struct
File:	Icu_17_GtmCcu6.h
Range:	Range : ICU_LOW_TIME : The channel is configured for reading the elapsed Signal Low Time ICU_HIGH_TIME : The channel is configured for reading the elapsed Signal High Time ICU_PERIOD_TIME : The channel is configured for reading the elapsed Signal Period Time ICU_ACTIVE_TIMEICU_DUTY_CYCLE: The channel is configured to read values which are needed for calculating the duty cycle (coherent Active and Period Time).
Description:	Definition of the measurement property type.ICU_DUTY_CYCLE is not provided according to ongoing Bugzilla 52843.

6.1.11 Type Definition Icu_17_GtmCcu6_TimestampBufferType

Table 75 Type Definition Icu TimestampBuffer Type

Syntax:	Icu_17_GtmCcu6_TimestampBufferType
Type:	Struct
File:	Icu_17_GtmCcu6.h
Range:	Range : ICU_LINEAR_BUFFER : The buffer will just be filled once ICU_CIRCULAR_BUFFER : After reaching the end of the buffer, the driver restarts at the beginning of the buffer
Description:	Definition of the timestamp measurement property type

6.1.12 Type Definition Icu_17_GtmCcu6_ClockType

Table 76 Type Definition Icu_17_GtmCcu6_MapType

Syntax:	Icu_17_GtmCcu6_MapType
Type:	uint8
File:	Icu_17_GtmCcu6.h
Range:	0 to 8 (GTM_CONFIGURABLE_CLOCK_0 to GTM_CONFIGURABLE_CLOCK_8)
Description:	Clock source of GTM channel.

6.1.13 Type Definition Icu_17_GtmCcu6_WakeupCapableType

Table 77 Type Definition Icu_17_GtmCcu6_WakeupCapableType

Syntax:	Icu_17_GtmCcu6_WakeupCapableType
Type:	uint8

File:	Icu_17_GtmCcu6.h	
Range:	ICU_NOT_WAKEUPCAPABLE	False: channel cannot be used for wakeup events.
	ICU_WAKEUPCAPABLE	True: channel can be used for wakeup events.
Description:	This type can be utilized while setting the ICU channel wakeup capability during configuration.	

6.1.14 Type Definition Icu_17_GtmCcu6_ChannelConfigType

Table 78 Type Definition Icu_17_GtmCcu6_ChannelConfigType

Syntax:	Icu_17_GtmCcu6_ChannelConfigType	
Type:	Struct	
File:	Icu_17_GtmCcu6.h	
Range:	IcuNotificationPointer (function pointer)	Callback notification function used in ICU_MODE_SIGNAL_EDGE_DETECT and ICU_MODE_TIMESTAMP
	IcuMeasurementMode (unsigned_int)	Measurement mode of the channel.
	IcuDefaultStartEdge (unsigned_int)	Start edge of the channel (default). Refer
	IcuMeasurementProperty (unsigned_int)	Signal measurement mode. Applicable if measurement mode is signal measurement.
	IcuWakeupCapability (unsigned_int)	TRUE, FALSE
	IcuAssignedHwUnit (unsigned_int)	HW resource for the channel. TIM Channel/ ERU Channel/CCU channel..
	IcuAssignedHwModule(unsigned_int)	Assigned GTM Module, ERU Module or CCU module.
	Ccu6Connections	CCU6 hardware connections.
	SignalType	ASIL or QM signal
	Reserved	Reserved for future use
	BufferMarker (uint32)	Buffer marker for time stamp capture and signal measurement. The size will be uint32. (Applicable if Safety is ON)
	TimChFilterTimeForRisingEdge (uint32)	Tim channel filter parameter for rising edge. The size will be uint16.
	TimChFilterTimeForFallingEdge (uint32)	Tim channel filter parameter for falling edge. The size will be uint16.
	TimInterruptMode (uint16)	Tim channel IRQ mode configuration value. The size will be uint16.
	TimChannelClockSelect (Icu_17_GtmCcu6_ClockType)	Tim channel clock select. The size will be of type Icu_17_GtmCcu6_ClockType.

	TimChannelFilterEnable (uint8)	Enables Filter for the Channel
	TimChFilterCounterFreqSelect (uint8)	Decides the filter Counter frequency for the TIM Channel
	TimChFilterModeForRisingEdge (uint8)	Decides the filter mode for rising Edge of the TIM Channel Input
	TimChFilterModeForFallingEdge (uint8)	Decides the filter mode for falling Edge of the TIM Channel Input
Description:	This type definition is used to configure ICU channel configuration.	

6.1.15 Type Definition Icu_17_GtmCcu6_MapType

Table 79 Type Definition Icu_17_GtmCcu6_MapType

Syntax:	Icu_17_GtmCcu6_MapType	
Type:	Structure	
File:	Icu_17_GtmCcu6.h	
Range:	PhysicalChannelMap	Channel number based on configured channel signal type (ASIL/QM)
	ChannelIndexMap	channel indexing for optimized RAM usage
Description:	The configured channel are divided into three groups ASIL-GTM, ASIL-CCU6 and QM. The PhysicalChannelMap maps the configured channel order to either of the three sets ASIL-GTM, ASIL-CCU6 or QM channels(GTM/CCU6/ERU)	

6.1.16 Type Definition Icu_17_GtmCcu6_ConfigType

Table 80 Type Definition Icu_17_GtmCcu6_ConfigType

Syntax:	Icu_17_GtmCcu6_ConfigType	
Type:	Struct	
File:	Icu_17_GtmCcu6.h	
	Marker	The marker holds the Icu module id and instance id. The size of the marker will be uint32.
	IcuChannelConfigPtr (Icu_17_GtmCcu6_ChannelConfigType)	Pointer to channel configuration set. (Valid FLASH address)
	IcuWakeupPtr (Icu_ChannelWakeupType)	Pointer to channel wakeup information. (Valid FLASH address)
	MappingIndexPtr	Pointer to Mapping of configuration channel numbers. To be used based on safety / QM usage and indexing for RAM optimization
	IcuErulntrChannelMap (uint8 [])	Interrupt map between ERU resources and ICU channels. For other resources (TIM Channel) the channel-interrupt map is placed with GTM driver.

	T12Parameters (uint8 [ICU_NO_OF_CCU6_MODULES])	Provides information about CCU6x kernel. The structure is the following, 1. T12ClkSelection -> Lower 3 bits (Bits 0 to 2) 2. T12PrescalerEnabled -> Bit 3 3. Wakeup Required -> Bit 4 4. Initialized by ICU or not -> Bit 7 ----- Initialization by ICU -> 0, ----- Other Driver Initialization or Unused -> 1
	IcuMaxChannels (uint8 [])	Array of 4 elements, each element is configured with the number of channels configured for each measurement mode.
Description:	This type definition is used to configure over all ICU configuration. The Pointer to the object of this type is used in Icu_17_GtmCcu6_Init API to initialize all the required resources.	

6.1.17 Type Definition Icu_17_GtmCcu6_MapType

Table 81 Type Definition Icu_17_GtmCcu6_MapType

Syntax:	Icu_17_GtmCcu6_MapType	
Type:	Structure	
File:	Icu_17_GtmCcu6.h	
Range:	PhysicalChannelMap	Channel number based on configured channel signal type (ASIL/QM)
	ChannelIndexMap	channel indexing for optimized RAM usage
Description:	The configured channel are divided into three groups ASIL-GTM, ASIL-CCU6 and QM. The PhysicalChannelMap maps the configured channel order to either of the three sets	

6.1.18 Type Definition Icu_17_GtmCcu6_ClockType

Table 82 Type Definition Icu_17_GtmCcu6_MapType

Syntax:	Icu_17_GtmCcu6_MapType	
Type:	uint8	
File:	Icu_17_GtmCcu6.h	
Range:	0 to 8 (GTM_CONFIGURABLE_CLOCK_0 to GTM_CONFIGURABLE_CLOCK_8)	
Description:	Clock source of GTM channel.	

6.1.19 Type Definition Icu_17_GtmCcu6_QmType

Table 83 Type Definition Icu_17_GtmCcu6_QmType

Syntax:	Icu_17_GtmCcu6_QmType	
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Type:	Structure	
File:	Icu_17_GtmCcu6_Local.h	
Range:	GlbIndexStatus [ICU_MAX_QM_CHANNELS] (uint32)	Array to store the kind of channel, configured edges, notification status, wakeup status, channel index for that measurement mode etc.
	MeasurementParas [ICU_MAX_SIGNAL_MEASURE_QM_CHANNELS]; (Icu_SigMeasurementParasType)	Array of maximum signal measurement channels, holds the run time parameters like capture timer values, the interrupt counters etc
	InfoPtr [ICU_MAX_SIGNAL_MEASURE_QM_CHANNELS]; (uint8)	Stores Mode specific information
	InfoPtrForTimeElapsed [ICU_MAX_SIGNAL_MEASURE_QM_CHANNELS] (uint8)	Stores Mode specific information
	TimeStampParas [ICU_MAX_TIMESTAMP_QM_CHANNELS] (Icu_TimeCaptureParasType)	Array of maximum timestamp measurement channels, holds the run time parameters like buffer size, counter to monitor whether size / notification is reached, notification interval.
	EdgeCountValuePtr [ICU_MAX_EDGE_COUNT_QM_CHANNELS] (uint32)	Stores the edge counter value at previous enable
	GtmEdgeCountForTimeElapsed [GTM_NO_OF_TIM_MODULES] (uint16)	Variable to store the Edge Counter overflow information
	GtmEdgeCountForInputState [GTM_NO_OF_TIM_MODULES] (uint16)	Variable to store the Edge Counter overflow information
	CcuActiveTime [ICU_NO_OF_CC6_MODULES] [ICU_NO_OF_CC6_CHANNELS] (Icu_17_GtmCcu6_ValueType)	Active time backup for CCU channels configured for Duty cycle
	CcuPeriodTime [ICU_NO_OF_CC6_MODULES] [ICU_NO_OF_CC6_CHANNELS] (Icu_17_GtmCcu6_ValueType)	Period time backup for CCU channels configured for Duty cycle
	CcuRRegVal [ICU_NO_OF_CC6_MODULES] [ICU_NO_OF_CC6_CHANNELS] (uint16)	R Register of CC6x backup value
	CcuSrRegVal [ICU_NO_OF_CC6_MODULES] [ICU_NO_OF_CC6_CHANNELS] (uint16)	SR Register of CC6x backup value

Description:	Structure to hold all QM(GTM,CCU6 and ERU) channels related Information of ICU driver.
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6.1.20 Type Definition Icu_17_GtmCcu6_AsilGtmType

Table 84 Type Definition Icu_17_GtmCcu6_AsilGtmType

Syntax:	Icu_17_GtmCcu6_AsilGtmType
Type:	Structure
File:	Icu_17_GtmCcu6_Local.h
Range:	Description of each parameter is same as Icu_17_GtmCcu6_QmType but for GTM ASIL channel. Refer Table 83 GlbIndexStatus[ICU_MAX_ASIL_GTM_CHANNELS] GtmEdgeCountForInputState[GTM_NO_OF_TIM_MODULES] GtmEdgeCountForTimeElapsed[GTM_NO_OF_TIM_MODULES] TimeStampParas[ICU_MAX_TIMESTAMP_ASIL_CHANNELS] EdgeCountValuePtr[ICU_MAX_EDGE_COUNT_ASIL_CHANNELS] MeasurementParas[ICU_MAX_SIGNAL_MEASURE_ASIL_GTM_CHANNELS] InfoPtr[ICU_MAX_SIGNAL_MEASURE_ASIL_GTM_CHANNELS] InfoPtrForTimeElapsed[ICU_MAX_SIGNAL_MEASURE_ASIL_GTM_CHANNELS]
Description:	Structure to hold all ASIL GTM channels related Information of ICU driver.

6.1.21 Type Definition Icu_17_GtmCcu6_AsilCcu6Type

Table 85 Type Definition Icu_17_GtmCcu6_AsilCcu6Type

Syntax:	Icu_17_GtmCcu6_AsilCcu6Type
Type:	Structure
File:	Icu_17_GtmCcu6_Local.h
Range:	Description of each parameter is same as Icu_17_GtmCcu6_QmType but for CCU6 ASIL channel. Refer Table 83 GlbIndexStatus[ICU_MAX_ASIL_CCU6_CHANNELS] MeasurementParas[ICU_MAX_SIGNAL_MEASURE_ASIL_CCU6_CHANNELS] CcuActiveTime[ICU_NO_OF_CCU6_MODULES][ICU_NO_OF_CCU6_CHANNELS] CcuActiveTime[ICU_NO_OF_CCU6_MODULES][ICU_NO_OF_CCU6_CHANNELS] CcuRRegVal[ICU_NO_OF_CCU6_MODULES][ICU_NO_OF_CCU6_CHANNELS] CcuSrRegVal[ICU_NO_OF_CCU6_MODULES][ICU_NO_OF_CCU6_CHANNELS]
Description:	Structure to hold all ASIL CCU6 channels related Information of ICU driver.

6.2 API Functions

This chapter describes the API interfaces provided by ICU Driver.

6.2.1 Icu_17_GtmCcu6_Init

Table 86 Specification of API Icu_17_GtmCcu6_Init

Service name:	Icu_17_GtmCcu6_Init	
Syntax:	<pre>void Icu_17_GtmCcu6_Init (const Icu_17_GtmCcu6_ConfigType *ConfigPtr)</pre>	
Service ID:	0x00	
Sync/ Async:	Synchronous	
Origin	Autosar	
Re-entrancy:	non reentrant	
Parameters (in):	ConfigPtr	Pointer to configuration set
Parameters (out):	None	
Return value:	None	
Description:	Service for ICU Driver initialization. The Initialization function initializes all relevant registers of the configured hardware with the values of the passed pointer to configuration set. Set the driver Init status to uninitialized if CCU6 Clock initialization is failed during the initialization and safety error will be reported in case of Safety is on.	
Caveats:	This service should not be called during a running operation. Icu_17_GtmCcu6_Init shall be invoked by the user only after PLL setup (Using Mcu_DistributePllClock).	
Configuration:	None	
DET:	ICU_E_PARAM_CONFIG	- Configuration Pointer is NULL. (If PBFixedAddress feature is OFF) - If PBFixedAddress feature is ON, DET will be raised if Icu_17_GtmCcu6_Init is called with any parameter other than Icu_ConfigRoot[0]
	ICU_E_ALREADY_INITIALIZED	Icu_17_GtmCcu6_Init called when ICU module is already initialized using a call of Icu_17_GtmCcu6_Init.
DEM:	None	
Implementation Comments:	a) If Icu_17_GtmCcu6_Init is called again before executing Icu_17_GtmCcu6_DeInit, then the application SW should take care of running channels / consequences. b) The following configurations are not valid (points c to e) c) ERU resource is configured for a measurement mode other than signal edge detection. d) Default Start Edge is configured as IcuBothEdges for IcuPeriodTime or <START AS40>IcuActiveTime<END AS40><START AS321>IcuDutyCycle<END AS321> measurement. e) Resource conflict (TIM/ERU) across channels or modules. The same resource is used in more than one channel. f) Gtm_Init should be called before executing Icu_17_GtmCcu6_Init. Gtm_Init will be called by the Mcu_Init of MCU Driver. g) In case of macro ICU_17_GTMCCU6_RESET_SFR_AT_INIT is STD_ON then the configured CCU6 and ERU registers are set to reset values before used in ICU driver. The GTM TIM SFRs will be reset by Gtm_Init if McuResetSfrAtInit is STD_ON in MCU configuration. This change will not impact the existing functionality of the driver.	
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided	

	configuration parameter IcuRunningInUser0Mode is 'True' and IcuUserModeInitApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeInitApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeInitApiEnable is 'False'.
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6.2.2 Icu_17_GtmCcu6_DeInit

Table 87 Specification of API Icu_17_GtmCcu6_DeInit

Service name:	Icu_17_GtmCcu6_DeInit
Syntax:	void Icu_17_GtmCcu6_DeInit(void)
Service ID:	0x01
Sync/ Async:	Synchronous
Origin	Autosar
Re-entrancy:	Non re-entrant
Parameters (in):	None
Parameters (out):	None
Return value:	None
Description:	Service for ICU De-Initialization. This service will disable all used interrupts and notifications. This service clears all pending interrupt flags. This brings the used resources (GTM/ERU/CCU6) which were modified in Icu_17_GtmCcu6_Init, back to their Power On Reset values. Non Writable registers are excluded.
Caveats:	None
Configuration:	IcuDeInitApi Table 21
DET:	ICU_E_UNINIT: ICU Driver is not initialized.
DEM:	None
Implementation Comments:	None
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter IcuRunningInUser0Mode is 'True' and IcuUserModeDeInitApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeDeInitApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeDeInitApiEnable is 'False'.

6.2.3 Icu_17_GtmCcu6_SetMode

Table 88 Specification of API Icu_17_GtmCcu6_SetMode

Service name:	Icu_17_GtmCcu6_SetMode
Syntax:	void Icu_17_GtmCcu6_SetMode (

	Icu_17_GtmCcu6_ModeType Mode)	
Service ID:	0x02	
Sync/ Async:	Synchronous	
Origin	Autosar	
Re-entrancy:	Non re-entrant	
Parameters (in):	Mode	ICU_MODE_NORMAL: All used interrupts are enabled according to notification requests. ICU_MODE_SLEEP: Reduced Power Mode, In sleep mode only those notifications are available which are configured as wakeup capable.
Parameters (out):	None	
Return value:	None	
Description:	If Requested mode is SLEEP, all the non wakeup capable channels will be disabled. If NORMAL mode is requested, all the non wakeup capable edge detection channels will be Enabled.	
Caveats:	None	
Configuration:	IcuSetModeApi	
DET:	ICU_E_UNINIT	Icu_17_GtmCcu6_Init not executed
	ICU_E_BUSY_OPERATION	if any measurement activity is not stopped then the API shall return busy
	ICU_E_PARAM_MODE	Input Mode is neither ICU_MODE_SLEEP nor ICU_MODE_NORMAL
DEM:	None.	
Implementation Comments:	None	
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter IcuRunningInUser0Mode is 'True' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'False'.	

6.2.4 Icu_17_GtmCcu6_DisableWakeup

Table 89 Specification of API Icu_17_GtmCcu6_DisableWakeup

Service name:	Icu_17_GtmCcu6_DisableWakeup
Syntax:	<pre>void Icu_17_GtmCcu6_DisableWakeup (Icu_17_GtmCcu6_ChannelType Channel)</pre>
Service ID:	0x03

Sync/ Async:	Synchronous	
Origin	Autosar	
Re-entrancy:	Re-entrant	
Parameters (in):	Channel	Numeric identifier of the channel.
Parameters (out):	None	
Return value:	None	
Description:	This service disables the wake-up capability of a single ICU channel for the following ICU mode selection(s). This service is only feasible for ICU channels configured as wake-up capable true. To make the selection effective a call of the function <code>Icu_17_GtmCcu6_SetMode</code>, requesting the mode to <code>ICU_MODE_SLEEP</code> is required.	
Caveats:	Usage of re-entrant capability is only allowed if the callers take care that there is no simultaneous usage of the same channel.	
Configuration:	IcuDisableWakeupApi Table 23	
DET:	ICU_E_UNINIT	ICU driver is not initialized.
	ICU_E_PARAM_CHANNEL	1. Invalid channel number. (Channel >= MaxChannels In the Configuration) 2. Channel is not having wakeup capability 3. Configured IcuMeasurementMode is not IcuSignalEdgeDetection
DEM:	None.	
Implementation Comments:	After <code>Icu_17_GtmCcu6_Init</code> the notifications are disabled by default. <code>Icu_17_GtmCcu6_EnableWakeup</code> should be called first to enable the wakeup of a channel then <code>Icu_17_GtmCcu6_DisableWakeup</code> API can perform accordingly. The wakeup information is used in <code>Icu_17_GtmCcu6_SetMode</code> API.	
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter <code>IcuRunningInUser0Mode</code> is 'True' and <code>IcuUserModeRuntimeApiEnable</code> is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter <code>IcuRunningInUser0Mode</code> is 'False' and <code>IcuUserModeRuntimeApiEnable</code> is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter <code>IcuRunningInUser0Mode</code> is 'False' and <code>IcuUserModeRuntimeApiEnable</code> is 'False'.	

6.2.5 Icu_17_GtmCcu6_EnableWakeup

Table 90 Specification for API Icu_17_GtmCcu6_EnableWakeup

Service name:	Icu_17_GtmCcu6_EnableWakeup	
Syntax:	void Icu_17_GtmCcu6_EnableWakeup(Icu_17_GtmCcu6_ChannelType Channel)	
Service ID:	0x04	
Sync/ Async:	Synchronous	
Origin	Autosar	
Re-entrancy:	Re-entrant	
Parameters (in):	Channel	Numeric identifier of the ICU channel

Parameters (out):	None	
Return value:	None	
Description:	This service enables / re-enables the wake-up capability of a single ICU channel for the following ICU mode selection(s). This service is only feasible for ICU channels configured as IcuWakeupCapability to true. To make the selection effective a call of the function Icu_17_GtmCcu6_SetMode, requesting the mode ICU_MODE_SLEEP is required.	
Caveats:	Usage of re-entrant capability is only allowed if the callers take care that there is no simultaneous usage of the same channel.	
Configuration:	IcuEnableWakeupApi Table 24	
DET:	ICU_E_UNINIT	ICU driver is not initialized.
	ICU_E_PARAM_CHANNEL	1. Invalid channel number. (Channel >= MaxChannels In the Configuration) 2. Channel is not having wakeup capability 3. Configured IcuMeasurementMode is not IcuSignalEdgeDetection.
DEM:	None.	
Implementation Comments:	None.	
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter IcuRunningInUser0Mode is 'True' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'False'.	

6.2.6 Icu_17_GtmCcu6_SetActivationCondition

Table 91 Specification of API Icu_17_GtmCcu6_SetActivationCondition

Service name:	Icu_17_GtmCcu6_SetActivationCondition	
Syntax:	<pre>void Icu_17_GtmCcu6_SetActivationCondition (Icu_17_GtmCcu6_ChannelType Channel Icu_17_GtmCcu6_Activationtype Activation)</pre>	
Service ID:	0x05	
Sync/ Async:	Synchronous	
Origin	Autosar	
Re-entrancy:	Re-entrant	
Parameters (in):	Channel	Numeric identifier of the ICU channel
	Activation	Edge (ICU_RISING_EDGE, ICU_FALLING_EDGE, ICU_BOTH_EDGES)
Parameters (out):	None	
Return value:	None	
Description:	This service sets the activation edge for the required measurement. This will override the DefaultStartEdge given in the configuration structure. However this	

	service is valid only when the Icu_17_GtmCcu6_MeasurementModeType is not ICU_MODE_SIGNAL_MEASUREMENT.	
Caveats:	Usage of re-entrant capability is only allowed if the callers take care that there is no simultaneous usage of the same channel.	
Configuration:	None	
DET:	ICU_E_UNINIT	ICU driver is not initialized.
	ICU_E_PARAM_CHANNEL	1. Invalid channel number. (Channel >= MaxChannels In the Configuration) 2. Configured IcuMeasurementMode is IcuSignalMeasurement.
	ICU_E_PARAM_ACTIVATION	Activation Edge is not equal to the following values ICU_RISING_EDGE, ICU_FALLING_EDGE, ICU_BOTH_EDGES.
DEM:	None.	
Implementation Comments:		
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter IcuRunningInUser0Mode is 'True' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'False'.	

6.2.7 Icu_17_GtmCcu6_DisableNotification

Table 92 Specification of API Icu_17_GtmCcu6_DisableNotification

Service name:	Icu_17_GtmCcu6_DisableNotification	
Syntax:	void Icu_17_GtmCcu6_DisableNotification (Icu_17_GtmCcu6_ChannelType Channel)	
Service ID:	0x06	
Sync/ Async:	Synchronous	
Origin	Autosar	
Re-entrancy:	Re-entrant	
Parameters (in):	Channel	Numeric identifier of the ICU channel
Parameters (out):	None	
Return value:	None	
Description:	This service disables all ICU signal notifications of this channel.	
Caveats:	Usage of re-entrant capability is only allowed if the callers take care that there is no simultaneous usage of the same channel.	
Configuration:	None	
DET:	ICU_E_UNINIT	ICU Driver is not initialized.
	ICU_E_PARAM_CHANNEL	Invalid channel number. (Channel >= MaxChannels In the

		Configuration) Configured IcuMeasurementMode is neither IcuTimestamp nor IcuSignalEdgeDetection
DEM:	None.	
Implementation Comments:	None.	
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter IcuRunningInUser0Mode is 'True' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'False'.	

6.2.8 Icu_17_GtmCcu6_EnableNotification

Table 93 Specification of API Icu_17_GtmCcu6_EnableNotification

Service name:	Icu_17_GtmCcu6_EnableNotification	
Syntax:	<pre>void Icu_17_GtmCcu6_EnableNotification(Icu_17_GtmCcu6_ChannelType Channel)</pre>	
Service ID:	0x07	
Sync/ Async:	Synchronous	
Origin	Autosar	
Re-entrancy:	Re-entrant	
Parameters (in):	Channel	Numeric identifier of the ICU channel
Parameters (out):	None	
Return value:	None	
Description:	This service enables the ICU signal notification according to either default start edge in Configuration or by the passed parameter given in Icu_17_GtmCcu6_SetActivationCondition (whichever is later). Timestamp Notification is also enabled through this API.	
Caveats:	Usage of re-entrant capability is only allowed if the callers take care that there is no simultaneous usage of the same channel.	
Configuration:	None	
DET:	ICU_E_UNINIT	ICU Driver is not initialized.
	ICU_E_PARAM_CHANNEL	Invalid channel number. (Channel >= MaxChannels In the Configuration) Configured IcuMeasurementMode is neither IcuTimestamp nor IcuSignalEdgeDetection
DEM:	None.	
Implementation Comments:	For timestamp notification, the user is expected to enable the notification then start the measurement by calling Icu_17_GtmCcu6_StartTimestamp. The other way round might result in loss of some interrupts, as this API cancels the pending interrupts.	

I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter IcuRunningInUser0Mode is 'True' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'False'.
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6.2.9 Icu_17_GtmCcu6_DisableEdgeDetection

Table 94 Specification of API Icu_17_GtmCcu6_DisableEdgeDetection

Service name:	Icu_17_GtmCcu6_DisableEdgeDetection	
Syntax:	void Icu_17_GtmCcu6_DisableEdgeDetection (Icu_17_GtmCcu6_ChannelType Channel)	
Service ID:	0x17	
Sync/ Async:	Synchronous	
Origin	Autosar	
Re-entrancy:	Re-entrant	
Parameters (in):	Channel	Numeric identifier of the ICU channel
Parameters (out):	None	
Return value:	None	
Description:	This service will disable the ICU signal edge detection channel. This is applicable only for channels with measurement mode of type Signal edge detection	
Caveats:	Usage of re-entrant capability is only allowed if the callers take care that there is no simultaneous usage of the same channel.	
Configuration:	IcuEdgeDetectApi Table 31	
DET:	ICU_E_UNINIT	ICU Driver is not initialized.
	ICU_E_PARAM_CHANNEL	Invalid channel number. (Channel >= MaxChannels In the Configuration) Configured IcuMeasurementMode is not IcuSignalEdgeDetection
DEM:	None.	
Implementation Comments:	None.	
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter IcuRunningInUser0Mode is 'True' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'False'.	

6.2.10 Icu_17_GtmCcu6_EnableEdgeDetection

Table 95 Specification of API Icu_17_GtmCcu6_EnableEdgeDetection

Service name:	Icu_17_GtmCcu6_EnableEdgeDetection	
Syntax:	void Icu_17_GtmCcu6_EnableEdgeDetection (Icu_17_GtmCcu6_ChannelType Channel)	
Service ID:	0x16	
Sync/ Async:	Synchronous	
Origin	Autosar	
Re-entrancy:	Re-entrant	
Parameters (in):	Channel	Numeric identifier of the ICU channel
Parameters (out):	None	
Return value:	None	
Description:	This service will Re enables the ICU signal edge detection channel, activation edge depends on either IcuDefaultStartEdge Or activation by Icu_17_GtmCcu6_SetActivationCondition API.	
Caveats:	Usage of re-entrant capability is only allowed if the callers take care that there is no simultaneous usage of the same channel.	
Configuration:	IcuEdgeDetectApi Table 31	
DET:	ICU_E_UNINIT	ICU Driver is not initialized.
	ICU_E_PARAM_CHANNEL	Invalid channel number. (Channel >= MaxChannels In the Configuration) Configured IcuMeasurementMode is not IcuSignalEdgeDetection
DEM:	None.	
Implementation Comments:	None.	
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter IcuRunningInUser0Mode is 'True' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'False'.	

6.2.11 Icu_17_GtmCcu6_GetInputState

Table 96 Specification of API Icu_17_GtmCcu6_GetInputState

Service name:	Icu_17_GtmCcu6_GetInputState	
Syntax:	Icu_17_GtmCcu6_InputStateType Icu_17_GtmCcu6_GetInputState (Icu_17_GtmCcu6_ChannelType Channel)	
Service ID:	0x08	
Sync/ Async:	Synchronous	

Origin	Autosar	
Re-entrancy:	Re-entrant	
Parameters (in):	Channel	Numeric identifier of the ICU channel
Parameters (out):	None	
Return value:	Icu_17_GtmCcu6_InputStateType Table 25	
Description:	This service returns the status of the ICU input. ICU_ACTIVE is returned for IcuSignalEdgeDetection channels if the configured edge has occurred. ICU_ACTIVE is returned for IcuSignalMeasurement channels if new data is available. The service sets the status to ICU_IDLE if there is no new value since the previous call of Icu_17_GtmCcu6_GetInputState, Icu_17_GtmCcu6_Init, Icu_17_GtmCcu6_EnableEdgeDetection(in case of Signal Edge Detection) or Icu_17_GtmCcu6_StartSignalMeasurement(in case of Signal Measurement)	
Caveats:	Usage of re-entrant capability is only allowed if the callers take care that there is no simultaneous usage of the same channel.	
Configuration:	None	
DET:	ICU_E_UNINIT	ICU Driver is not initialized.
	ICU_E_PARAM_CHANNEL	Invalid channel number. (Channel >= MaxChannels In the Configuration) If Channel is not configured as ICU_MODE_SIGNAL_EDGE_DETECT Or ICU_MODE_SIGNAL_MEASUREMENT
DEM:	None.	
Implementation Comments:	None	
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter IcuRunningInUser0Mode is 'True' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'False'.	

6.2.12 Icu_17_GtmCcu6_StartTimestamp

Table 97 Specification of API Icu_17_GtmCcu6_StartTimestamp

Service name:	Icu_17_GtmCcu6_StartTimestamp
Syntax:	<pre>void Icu_17_GtmCcu6_StartTimestamp (Icu_17_GtmCcu6_ChannelType Channel, Icu_17_GtmCcu6_ValueType *BufferPtr, uint16 BufferSize, uint16 NotifyInterval)</pre>
Service ID:	0x09

Sync/ Async:	Asynchronous	
Origin	Autosar	
Re-entrancy:	Re-entrant	
Parameters (in):	Channel	Numeric identifier of the channel
	BufferSize	Size of the Buffer
	NotifyInterval	Number of Timestamp events on which a notification will be called.
Parameters (out):	BufferPtr	Pointer to the buffer where the timestamp values shall be placed.
Return value:	None	
Description:	This service starts the Timestamp capture. The actual capture will happen in the ISR. Notifications will also be called from the ISR only. If safety is ON and invoked for an ASIL channel, application should fill BufferPtr[0] with the value configured for parameter IcuTimestampBufferMarker. If safety is ON and invoked for a QM channel, application should fill BufferPtr[0] with zero.	
Caveats:	Usage of re-entrant capability is only allowed if the callers take care that there is no simultaneous usage of the same channel.	
Configuration:	IcuTimestampApi Table 29	
DET:	ICU_E_UNINIT	ICU Driver is not initialized.
	ICU_E_PARAM_CHANNEL	Invalid channel number. (Channel >= MaxChannels In the Configuration) Configured IcuMeasurementMode is not IcuTimestamp.
	ICU_E_PARAM_BUFFER_PTR	Buffer Pointer is 0
	ICU_E_PARAM_BUFFER_SIZE	Buffer Size is 0
	ICU_E_PARAM_NOTIFY_INTERVAL	Notify Interval is invalid (i.e. "0")
DEM:	None.	
Implementation Comments:	None	
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter IcuRunningInUser0Mode is 'True' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'False'.	

6.2.13 Icu_17_GtmCcu6_StopTimestamp

Table 98 Specification of API Icu_17_GtmCcu6_StopTimestamp

Service name:	Icu_17_GtmCcu6_StopTimestamp
Syntax:	void Icu_17_GtmCcu6_StopTimestamp

	(Icu_17_GtmCcu6_ChannelType Channel)	
Service ID:	0x0A	
Sync/ Async:	Synchronous	
Origin	Autosar	
Re-entrancy:	Re-entrant	
Parameters (in):	Channel	Numeric identifier of the channel
Parameters (out):	None	
Return value:	None	
Description:	This service stops the timestamp capture of the particular channel.	
Caveats:	Usage of re-entrant capability is only allowed if the callers take care that there is no simultaneous usage of the same channel.	
Configuration:	IcuTimestampApi Table 29	
DET:	ICU_E_UNINIT	ICU driver is not initialized.
	ICU_E_NOT_STARTED	The Measurement is not started.
	ICU_E_PARAM_CHANNEL	Invalid channel number. (Channel >= MaxChannels In the Configuration) Configured IcuMeasurementMode is not IcuTimestamp.
DEM:	None.	
Implementation Comments:	None	
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter IcuRunningInUser0Mode is 'True' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'False'.	

6.2.14 Icu_17_GtmCcu6_GetTimestampIndex

Table 99 Specification of API Icu_17_GtmCcu6_GetTimestampIndex

Service name:	Icu_17_GtmCcu6_GetTimestampIndex	
Syntax:	Icu_17_GtmCcu6_IndexType Icu_17_GtmCcu6_GetTimestampIndex (Icu_17_GtmCcu6_ChannelType Channel)	
Service ID:	0x0B	
Sync/ Async:	Synchronous	
Origin	Autosar	
Re-entrancy:	Re-entrant	
Parameters (in):	Channel	Numeric identifier of the channel

Parameters (out):	None	
Return value:	Icu_17_GtmCcu6_IndexType	
Description:	This service reads the Timestamp Index of the given channel that would be used when the next ISR occurs. A counter is maintained in a structure, which will be incremented after ISR. The counter will be returned in this API. If the API is called before executing Icu_17_GtmCcu6_StartTimestamp then the API returns zero.	
Caveats:	Usage of re-entrant capability is only allowed if the callers take care that there is no simultaneous usage of the same channel.	
Configuration:	IcuTimestampApi Table 29	
DET:	ICU_E_UNINIT	ICU Driver is not initialized.
	ICU_E_PARAM_CHANNEL	Invalid channel number. (Channel >= MaxChannels In the Configuration) Configured IcuMeasurementMode is not IcuTimestamp.
DEM:	None.	
Implementation Comments:	None	
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter IcuRunningInUser0Mode is 'True' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'False'.	

6.2.15 Icu_17_GtmCcu6_ResetEdgeCount

Table 100 Specification of API Icu_17_GtmCcu6_ResetEdgeCount

Service name:	Icu_17_GtmCcu6_ResetEdgeCount	
Syntax:	<pre>Icu_17_GtmCcu6_IndexType Icu_17_GtmCcu6_ResetEdgeCount (Icu_17_GtmCcu6_ChannelType Channel)</pre>	
Service ID:	0x0C	
Sync/ Async:	Synchronous	
Origin	Autosar	
Re-entrancy:	Re-entrant	
Parameters (in):	Channel	Numeric identifier of the channel
Parameters (out):	None	
Return value:	None	
Description:	The value of the counted edges will be zero with the call of this service.	
Caveats:	Usage of re-entrant capability is only allowed if the callers take care that there is no simultaneous usage of the same channel.	
Configuration:	IcuEdgeCountApi Table 30	
DET:	ICU_E_UNINIT	ICU Driver is not initialized.

	ICU_E_PARAM_CHANNEL	Invalid channel number. (Channel >= MaxChannels In the Configuration) The configured measurement mode is not IcuEdgeCounter.
DEM:	None.	
Implementation Comments:	None	
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter IcuRunningInUser0Mode is 'True' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'False'.	

6.2.16 Icu_17_GtmCcu6_EnableEdgeCount

Table 101 Specification of API Icu_17_GtmCcu6_EnableEdgeCount

Service name:	Icu_17_GtmCcu6_EnableEdgeCount	
Syntax:	<pre>void Icu_17_GtmCcu6_EnableEdgeCount (Icu_17_GtmCcu6_ChannelType Channel)</pre>	
Service ID:	0x0D	
Sync/ Async:	Synchronous	
Origin	Autosar	
Re-entrancy:	Re-entrant	
Parameters (in):	Channel	Numeric identifier of the channel
Parameters (out):	None	
Return value:	None	
Description:	The service enables counting of edges. This is applicable only for channels configured in Edge Count mode	
Caveats:	Usage of re-entrant capability is only allowed if the callers take care that there is no simultaneous usage of the same channel.	
Configuration:	IcuEdgeCountApi Table 30	
DET:	ICU_E_UNINIT	ICU Driver is not initialized.
	ICU_E_PARAM_CHANNEL	1. Invalid channel number. (Channel >= MaxChannels In the Configuration) 2. The configured measurement mode is not IcuEdgeCounter.
DEM:	None.	
Implementation Comments:	None.	
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter IcuRunningInUser0Mode is 'True' and	

	IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'False'.
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6.2.17 Icu_17_GtmCcu6_DisableEdgeCount

Table 102 Specification of API Icu_17_GtmCcu6_DisableEdgeCount

Service name:	Icu_17_GtmCcu6_DisableEdgeCount	
Syntax:	<pre>void Icu_17_GtmCcu6_DisableEdgeCount (Icu_17_GtmCcu6_ChannelType Channel)</pre>	
Service ID:	0x0E	
Sync/ Async:	Synchronous	
Origin	Autosar	
Re-entrancy:	Re-entrant	
Parameters (in):	Channel	Numeric identifier of the channel
Parameters (out):	None	
Return value:	None	
Description:	The service disables the edge count measurement of a channel that was started earlier. This is applicable only for channels configured in Edge Count mode	
Caveats:	Usage of re-entrant capability is only allowed if the callers take care that there is no simultaneous usage of the same channel.	
Configuration:	IcuEdgeCountApi Table 30	
DET:	ICU_E_UNINIT	ICU Driver is not initialized.
	ICU_E_PARAM_CHANNEL	Invalid channel number. (Channel >= MaxChannels In the Configuration) The configured measurement mode is not IcuEdgeCounter.
DEM:	None.	
Implementation Comments:	None	
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter IcuRunningInUser0Mode is 'True' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'False'.	

6.2.18 Icu_17_GtmCcu6_GetEdgeNumbers

Table 103 Specification of API Icu_17_GtmCcu6_GetEdgeNumbers

Service name:	Icu_17_GtmCcu6_GetEdgeNumbers	
Syntax:	Icu_17_GtmCcu6_EdgeNumberType Icu_17_GtmCcu6_GetEdgeNumbers (Icu_17_GtmCcu6_ChannelType Channel)	
Service ID:	0x0F	
Sync/ Async:	Synchronous	
Origin	Autosar	
Re-entrancy:	Re-entrant	
Parameters (in):	Channel	Numeric identifier of the channel
Parameters (out):	None	
Return value:	None	
Description:	The Service returns the no of edges counted	
Caveats:	Usage of re-entrant capability is only allowed if the callers take care that there is no simultaneous usage of the same channel.	
Configuration:	IcuEdgeCountApi Table 30	
DET:	ICU_E_UNINIT	ICU Driver is not initialized.
	ICU_E_PARAM_CHANNEL	Invalid channel number. (Channel >= MaxChannels In the Configuration) The configured measurement mode is not IcuEdgeCounter.
DEM:	None.	
Implementation Comments:	None	
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter IcuRunningInUser0Mode is 'True' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'False'.	

6.2.19 Icu_17_GtmCcu6_StartSignalMeasurement

Table 104 Specification of API Icu_17_GtmCcu6_StartSignalMeasurement

Service name:	Icu_17_GtmCcu6_StartSignalMeasurement	
Syntax:	void Icu_17_GtmCcu6_StartSignalMeasurement (Icu_17_GtmCcu6_ChannelType Channel)	
Service ID:	0x13	
Sync/ Async:	Asynchronous	

Origin	Autosar	
Re-entrancy:	Re-entrant	
Parameters (in):	Channel	Numeric identifier of the ICU channel
Parameters (out):	None	
Return value:	None	
Description:	This service starts the measurement of signals beginning with the configured default start edge which occurs first after the call of this service.	
Caveats:	Usage of re-entrant capability is only allowed if the callers take care that there is no simultaneous usage of the same channel.	
Configuration:	IcuSignalMeasurementApi	
DET:	ICU_E_UNINIT	ICU Driver is not initialized.
	ICU_E_PARAM_CHANNEL	If Channel is invalid (invalid identifier or channel not configured for mode ICU_MODE_SIGNAL_MEASUREMENT)
DEM:	None.	
Implementation Comments:		
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter IcuRunningInUser0Mode is 'True' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'False'.	

6.2.20 Icu_17_GtmCcu6_StopSignalMeasurement

Table 105 Specification of API Icu_17_GtmCcu6_StopSignalMeasurement

Service name:	Icu_17_GtmCcu6_StopSignalMeasurement	
Syntax:	<pre>void Icu_17_GtmCcu6_StopSignalMeasurement (Icu_17_GtmCcu6_ChannelType Channel)</pre>	
Service ID:	0x14	
Sync/ Async:	Synchronous	
Origin	Autosar	
Re-entrancy:	Re-entrant	
Parameters (in):	Channel	Numeric identifier of the channel
Parameters (out):	None	
Return value:	None	
Description:	This service stops the measurement of signals of the given channel.	
Caveats:	Usage of re-entrant capability is only allowed if the callers take care that there is no simultaneous usage of the same channel.	
Configuration:	IcuSignalMeasurementApi	
DET:	ICU_E_UNINIT	ICU Driver is not initialized.

	ICU_E_PARAM_CHANNEL	If Channel is invalid (invalid identifier or channel not configured for mode ICU_MODE_SIGNAL_MEASUREMENT)
DEM:	None.	
Implementation Comments:		
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter IcuRunningInUser0Mode is 'True' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'False'.	

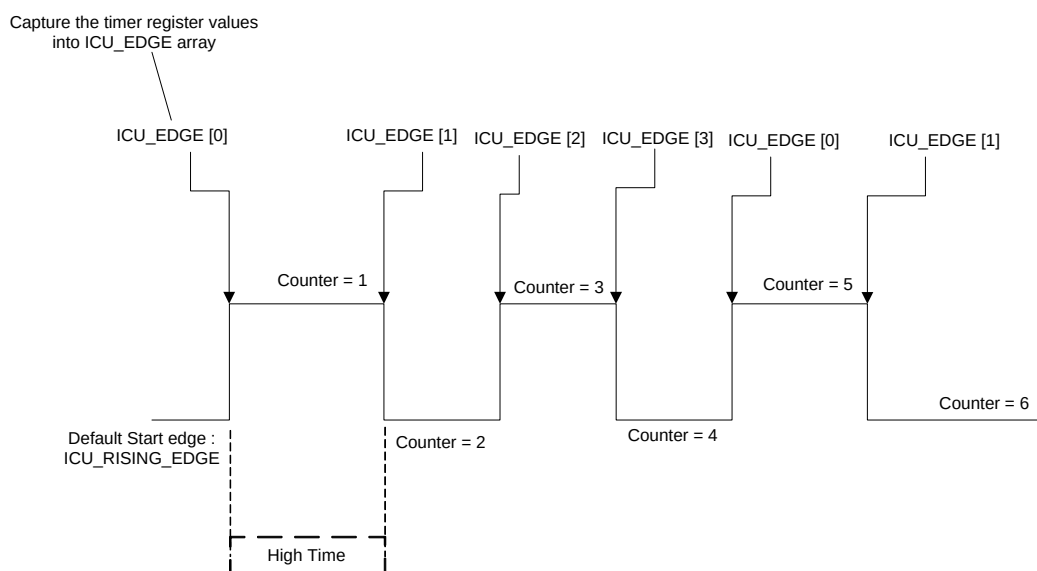
6.2.21 Icu_17_GtmCcu6_GetTimeElapsed

Table 106 Specification of API Icu_17_GtmCcu6_GetTimeElapsed

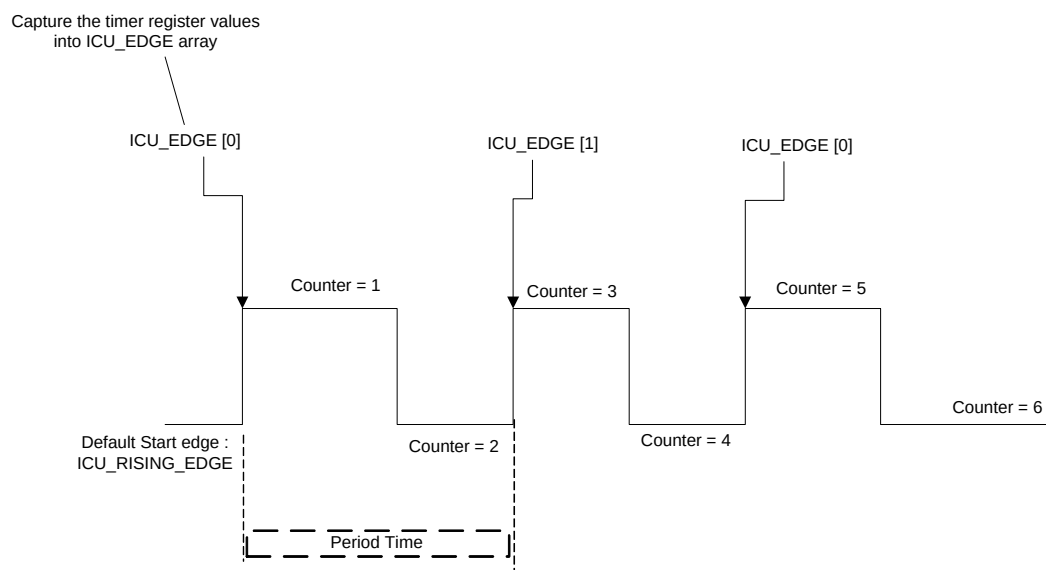
Service name:	Icu_17_GtmCcu6_GetTimeElapsed	
Syntax:	Icu_17_GtmCcu6_ValueType Icu_17_GtmCcu6_GetTimeElapsed (Icu_17_GtmCcu6_ChannelType Channel)	
Service ID:	0x10	
Sync/ Async:	Synchronous	
Origin	Autosar	
Re-entrancy:	Re-entrant	
Parameters (in):	Channel	Numeric identifier of the channel
Parameters (out):	None	
Return value:	None	
Description:	The service reads the elapsed high time / low time / period time for a particular channel as per configuration. If sufficient number of edges has not happened or new data is not available then the service returns 0.	
Caveats:	Usage of re-entrant capability is only allowed if the callers take care that there is no simultaneous usage of the same channel.	
Configuration:	IcuGetTimeElapsedApi Table 26	
DET:	ICU_E_UNINIT	ICU Driver is not initialized.
	ICU_E_PARAM_CHANNEL	Invalid channel number. (Channel >= MaxChannels In the Configuration) The configuration should be done for IcuSignalMeasurement.
DEM:	None.	
Implementation Comments:	Refer Figure 5 for Period, High and Low Time Measurement.	
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter IcuRunningInUser0Mode is 'True' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided	

configuration parameter `IcuRunningInUser0Mode` is 'False' and `IcuUserModeRuntimeApiEnable` is 'True'.
This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter `IcuRunningInUser0Mode` is 'False' and `IcuUserModeRuntimeApiEnable` is 'False'.

HIGH TIME MEASUREMENT



PERIOD TIME MEASUREMENT



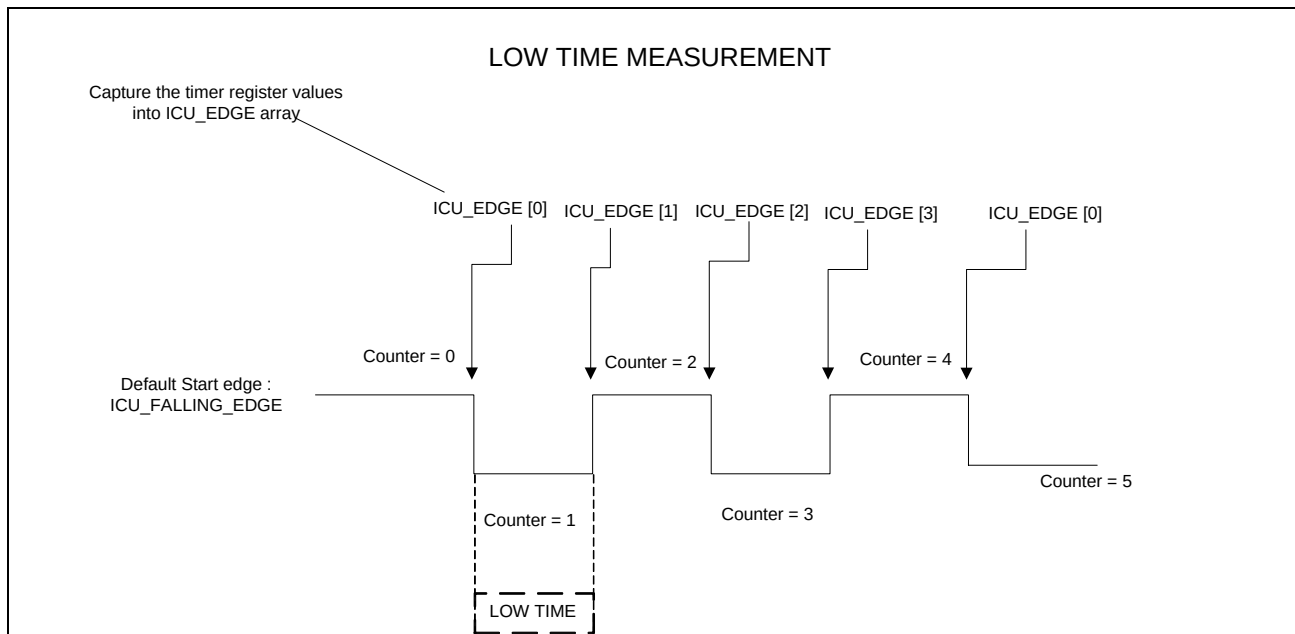


Figure 5 Counter and Time Capture for Period Time, High Time and Low Time

6.2.22 Icu_17_GtmCcu6_GetDutyCycleValues

Table 107 Specification of API Icu_17_GtmCcu6_GetDutyCycleValues

Service name:	Icu_17_GtmCcu6_GetDutyCycleValues	
Syntax:	<pre>Icu_17_GtmCcu6_ValueType Icu_17_GtmCcu6_GetDutyCycleValues (Icu_17_GtmCcu6_ChannelType Channel, Icu_17_GtmCcu6_DutyCycleType *DutyCycleValues)</pre>	
Service ID:	0x11	
Sync/ Async:	Synchronous	
Origin	Autosar	
Re-entrancy:	Re-entrant	
Parameters (in):	Channel	Numeric identifier of the channel
Parameters (out):	DutyCycleValues	Pointer to a buffer where the high time and period time shall be placed.
Return value:	None	
Description:	The service reads the elapsed duty cycle and writes it into the passed buffer pointer. If new data is not available then the service updates the buffer with 0. If safety is ON and invoked for an ASIL channel, application should fill the BufferMarker element of parameter DutyCycleValues with the value configured for parameter IcuDutycycleBufferMarker. If safety is ON and invoked for a QM channel, application should fill the BufferMarker element of parameter DutyCycleValues with zero.	
Caveats:	Usage of re-entrant capability is only allowed if the callers take care that there is no simultaneous usage of the same channel.	
Configuration:	IcuGetDutyCycleValuesApi	
DET:	ICU_E_UNINIT	ICU Driver is not initialized.

	ICU_E_PARAM_CHANNEL	Invalid channel number. (Channel >= MaxChannels In the Configuration) If Channel is not configured for IcuSignalMeasurement with The measurement property DutyCycle.
	ICU_E_PARAM_BUFFER_PTR	DutyCycleValues is invalid (0).
DEM:	None.	
Implementation Comments:	Refer Figure 6 for Duty Cycle Measurement.	
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter IcuRunningInUser0Mode is 'True' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'False'.	

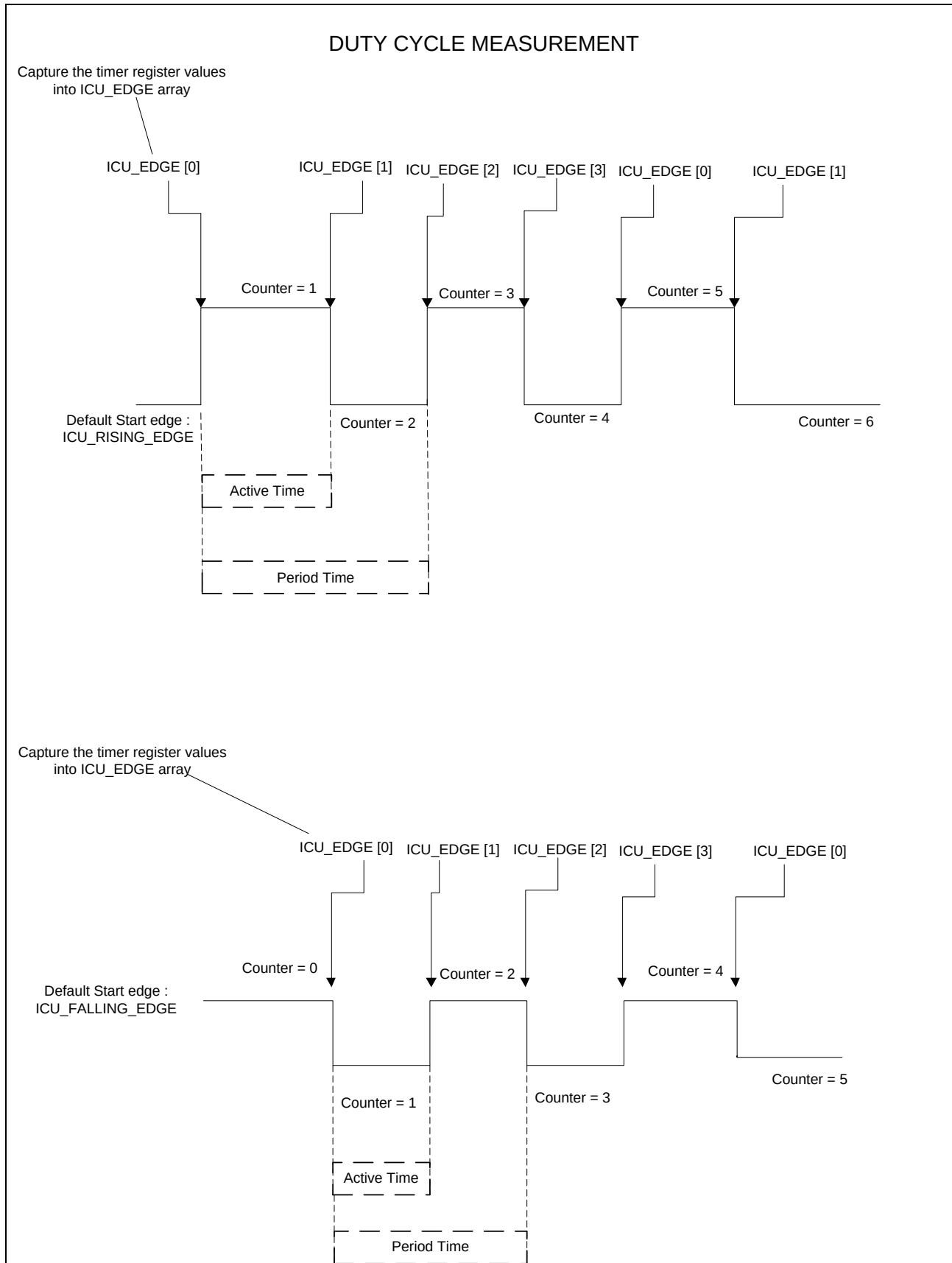


Figure 6 Counter and Time Capture for Duty Cycle Measurement

6.2.23 Icu_17_GtmCcu6_GetVersionInfo

Table 108 Specification of API Icu_17_GtmCcu6_GetVersionInfo

Service name:	Icu_17_GtmCcu6_GetVersionInfo	
Syntax:	<pre>void Icu_17_GtmCcu6_GetVersionInfo (Std_VersionInfoType *VersionInfoPtr)</pre>	
Service ID:	0x12	
Sync/ Async:	Synchronous	
Origin	Autosar	
Re-entrancy:	Non -reentrant	
Parameters (in):	None	
Parameters (out):	VersionInfoPtr	Pointer to valid memory space in order to store the version information of this module.
Return value:	None	
Description:	This service returns the version information of ICU module. This API is implemented as a Macro.	
Caveats:	None	
Configuration:	IcuGetVersionInfoApi Table 28	
DET:	ICU_E_PARAM_VINFO - VersionInfoPtr is passed as NULL_PTR	
DEM:	None.	
Implementation Comments:		
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter IcuRunningInUser0Mode is 'True' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'False'.	

6.2.24 Icu_17_GtmCcu6_CheckWakeup

Table 109 Functional Specification - Icu_17_GtmCcu6_CheckWakeup

Service name:	Icu_17_GtmCcu6_CheckWakeup	
Syntax:	<pre>void Icu_17_GtmCcu6_CheckWakeup (EcuM_WakeupSourceType wakeupSource)</pre>	
Service ID:	0x15	
Sync/Async:	Synchronous	
Origin	Autosar	
Reentrancy:	Reentrant	
Parameters (in):	wakeupSource	Channel wakeup info.

Parameters (out):	None	
Return value:	None	
Description:	Checks if a wakeup capable ICU channel is the source for a wakeup event and calls the ECU state manager service EcuM_SetWakeupEvent in case of a valid ICU channel wakeup event.	
Caveats:	Usage of Re-entrant capability is only allowed if the callers take care that there is no simultaneous usage of the same channel.	
Configuration:	This function is pre-compile time configurable STD_ON / STD_OFF by the configuration parameter IcuWakeupFunctionalityApi.	
DET:	ICU_E_UNINIT	If function is called prior to initialization (Init function).
DEM:	None	
Implementation Comments:	None	
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter IcuRunningInUser0Mode is 'True' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'False'.	

6.2.25 Icu_17_GtmCcu6_InitCheck

Table 110 Specification of API Icu_17_GtmCcu6_InitCheck

Service name:	Icu_17_GtmCcu6_InitCheck	
Syntax:	Std_ReturnType Icu_17_GtmCcu6_InitCheck(const Icu_17_GtmCcu6_ConfigType *ConfigPtr)	
Service ID:	None	
Sync/ Async:	Synchronous	
Origin	Vendor specific	
Re-entrancy:	Non re-entrant	
Parameters (in):	ConfigPtr : Pointer to configuration set	
Parameters (out):	None	
Return value:	Std_ReturnType	
Description:	Will check against all SFRs or variables initialized by Icu_17_GtmCcu6_Init including Icu_17_GtmCcu6_InitStatus. It does not modify any SFR/variable, only a read operation is done. If any failure in comparison, it reports an error.	
Caveats:	None	
Configuration:	Init check API is ON as well as Safety is ON	
DET:	None	
DEM:	None	
Implementation Comments:	This API needs to be executed only after Icu_17_GtmCcu6_Init().	
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided	

	configuration parameter IcuRunningInUser0Mode is 'True' and IcuUserModelInitApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModelInitApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModelInitApiEnable is 'False'.
--	--

6.2.26 Icu_17_GtmCcu6_GetMode

Table 111 Specification of API Icu_17_GtmCcu6_GetMode

Service name:	Icu_17_GtmCcu6_GetMode
Syntax:	Icu_17_GtmCcu6_ModeType Icu_17_GtmCcu6_GetMode(void)
Service ID:	None
Sync/ Async:	Synchronous
Origin	Vendor specific
Re-entrancy:	Non re-entrant
Parameters (in):	None
Parameters (out):	None
Return value:	Icu_17_GtmCcu6_ModeType
Description:	This service returns the mode of the ICU driver
Caveats:	None
Configuration:	This API is available only if ICU_SAFETY_ENABLE and ICU_GETMODE_API is STD_ON.
DET:	None.
DEM:	None.
Implementation Comments:	Return the value of global variable Icu_17_GtmCcu6_GlobMode holding the ICU driver mode.
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter IcuRunningInUser0Mode is 'True' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'False'.

6.2.27 Icu_17_GtmCcu6_EnableMultiEdgeDetection

Table 112 Specification of API Icu_17_GtmCcu6_EnableMultiEdgeDetection

Service name:	Icu_17_GtmCcu6_EnableMultiEdgeDetection
Syntax:	void Icu_17_GtmCcu6_EnableMultiEdgeDetection (Icu_17_GtmCcu6_ChannelType Channel, uint32 EdgeCount)
Service ID:	0x19
Sync/ Async:	Synchronous
Origin	Vendor specific

Re-entrancy:	Re-entrant	
Parameters (in):	Channel	Numeric identifier of the ICU channel
	EdgeCount	Number of edges before interrupt occurs
Parameters (out):	None	
Return value:	None	
Description:	- This service will generate the interrupt in ICU edge detection measurement mode after specified number of edges. - No extra configuration parameter is added to support the multi edge detection feature. It is considered as one of the part of edge detection mode. - The channels which are configured for edge detection mode will be used for multi edge detection feature. - By default after Icu_17_GtmCcu6_Init(), the edge detection mode is started and by calling Icu_17_GtmCcu6_EnableMultiEdgeDetection() the mode will be shifted from edge detection mode to multi edge detection mode. - To disable the multi edge detection mode, call Icu_17_GtmCcu6_DisableEdgeDetection() API.	
Caveats:	Usage of re-entrant capability is only allowed if the callers take care that there is no simultaneous usage of the same channel.	
Configuration:	Same as IcuEdgeDetectApi Table 31	
DET:	ICU_E_UNINIT	ICU Driver is not initialized.
	ICU_E_PARAM_CHANNEL	Invalid channel number. (Channel >= MaxChannels In the Configuration) Configured IcuMeasurementMode is not IcuSignalEdgeDetection
	ICU_E_PARAM_EDGE_NUMBER	Function called with a EdgeCount parameter as zero. In case of wrong EdgeCount, DET will occur and the MultiEdge detection feature will not be started but the previous edge detection mode will be continued for the channel.
DEM:	None.	
Implementation Comments:	None.	
I/O Mode	This API shall be invoked when the CPU is in User-0 Mode provided configuration parameter IcuRunningInUser0Mode is 'True' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in User-1 Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'True'. This API shall be invoked when the CPU is in Supervisor Mode provided configuration parameter IcuRunningInUser0Mode is 'False' and IcuUserModeRuntimeApiEnable is 'False'.	

6.3 Debug Macros

- There are debug code written within the driver to simulate certain HW conditions, however these code fragments are encapsulated with a DEBUG macro i.e. the final binary will not have these instructions. The macros are like IFX_ICU_DEBUG.

- The enable of macro `ICU_DEBUG_SUPPORT` provides the variables that are relevant for debugging purpose as Global variables.
- The debug macro `IFX_ICU_DEBUG01` is provided to simulate the error condition of CCU6 clock initialization, using variable `TestIcu_DebugMask01`. There is no definition of the variable done within the code and user has to define the same.

6.4 Interrupt Handling

The interrupt service routines play a major role in satisfying most of the ICU requirements.

General Notes:

- The ISR's are optimized according to the measurement mode used. `Icu_17_GtmCcu6_Isr` and `Icu_17_GtmCcu6_Eru_Isr` has the capability to handle all the measurement modes. The user should take care in assigning one resource per measurement mode..
- Each required SRN must be given a priority in the supplied `Irq` configuration and the corresponding SRN dependant resources should be enabled.
- Interrupt should be enabled for Channels configured for the following measurement modes:
 - Edge Detect Mode (if Notification is enabled for that channel if the resource used is TIM)
 - Edge Detect Mode (if resource used is ERU)
 - Edge Detect Mode (if resource used is CCU6)
 - Timestamp Mode
 - Signal Measurement Mode (If GTM is used, then interrupt is needed only in case of High Time or Low Time only)
 - Signal Measurement Mode (If CCU6 is used, then interrupt is needed for all modes)

The following table (Table 113) shows the SRN Number and the associated resource.

Table 113 Service Request Node – Hardware Resource

Hardware Resource		SRN Number
TIM0	CHANNEL0	SRC_GTMTIM00
	CHANNEL1	SRC_GTMTIM01
	CHANNEL2	SRC_GTMTIM02
	CHANNEL3	SRC_GTMTIM03
	CHANNEL4	SRC_GTMTIM04
	CHANNEL5	SRC_GTMTIM05
	CHANNEL6	SRC_GTMTIM06
	CHANNEL7	SRC_GTMTIM07
TIM1	CHANNEL0	SRC_GTMTIM10
	CHANNEL1	SRC_GTMTIM11
	CHANNEL2	SRC_GTMTIM12
	CHANNEL3	SRC_GTMTIM13
	CHANNEL4	SRC_GTMTIM14
	CHANNEL5	SRC_GTMTIM15
	CHANNEL6	SRC_GTMTIM16
	CHANNEL7	SRC_GTMTIM17
TIM2	CHANNEL0	SRC_GTMTIM20
	CHANNEL1	SRC_GTMTIM21
	CHANNEL2	SRC_GTMTIM22
	CHANNEL3	SRC_GTMTIM23
	CHANNEL4	SRC_GTMTIM24

Hardware Resource		SRN Number
	CHANNEL5	SRC_GTMTIM25
	CHANNEL6	SRC_GTMTIM26
	CHANNEL7	SRC_GTMTIM27
TIM3	CHANNEL0	SRC_GTMTIM30
	CHANNEL1	SRC_GTMTIM31
	CHANNEL2	SRC_GTMTIM32
	CHANNEL3	SRC_GTMTIM33
	CHANNEL4	SRC_GTMTIM34
	CHANNEL5	SRC_GTMTIM35
	CHANNEL6	SRC_GTMTIM36
	CHANNEL7	SRC_GTMTIM37
ERU	CHANNEL 0 to 7	SRC_SCUESR0
CCU60	CHANNEL CC60 to CC62	SRC_CCU60SR0
CCU61	CHANNEL CC60 to CC62	SRC_CCU61SR0

The following [Table 114](#) gives the summary of all interrupt service requests handled by ICU Driver.

Table 114 Interrupt Service Routines

Hardware Resource	Name of the Routine	Input Parameter	Description
Tim	Icu_17_GtmCcu6_Isr	Tim Channel Number	All the TIM Channels are mapped to this ISR. TIM can be used for all measurement modes.
ERU	Icu_17_GtmCcu6_Eru_Isr	Channel	The External Request Unit sources can be used only for signal edge detection / notification. ERU lines are handled in this ISR.
CCU6	Icu_17_GtmCcu6_Ccu_Isr	CCU6x Kernel number	All the CCU6x channels are mapped to this Isr.

6.4.1 Icu_17_GtmCcu6_Isr

Table 115 Specification of API Icu_17_GtmCcu6_Isr

Service name:	Icu_17_GtmCcu6_Isr	
Syntax:	Icu_17_GtmCcu6_Isr (uint8 Channel)	
Service ID:	0x18	
Sync/ Async:	Synchronous	
Origin	Vendor specific	
Re-entrancy:	Reentrant	
	ChannelNo	Icu Channel Number
Parameters (out):	None	
Return value:	None	
Description:	ISR to handle TIM Channel requests. As stated earlier TIM Channels can handle all possible measurement modes. The interrupt source check is done by GTM driver as Icu_17_GtmCcu6_Isr is called from GTM ISR.	

Caveats:	None
Configuration:	None
DET:	None
DEM:	None
Implementation Comments:	None
I/O Mode	This API can be invoked when the CPU is in either User-1 Mode or Supervisor Mode.

6.4.2 Icu_17_GtmCcu6_Eru_Isr

Table 116 Specification of API Icu_17_GtmCcu6_Eru_Isr

Service name:	Icu_17_GtmCcu6_Eru_Isr
Syntax:	void Icu_17_GtmCcu6_Eru_Isr (void)
Service ID:	0x1B
Sync/ Async:	Synchronous
Origin	Vendor specific
Re-entrancy:	Reentrant
Parameters (in):	None
Parameters (out):	None
Return value:	None
Description:	ISR to handle ERU requests. ERU resources can handle only edge detection mode. Report safety error ICU_E_INTR_SRC_ERROR in case of interrupt source is wrong.
Caveats:	None
Configuration:	None
DET:	None
DEM:	None
Implementation Comments:	None
I/O Mode	This API can be invoked when the CPU is in either User-1 Mode or Supervisor Mode.

6.4.3 Icu_17_GtmCcu6_Ccu_Isr

Table 117 Specification of API Icu_17_GtmCcu6_Ccu_Isr

Service name:	Icu_17_GtmCcu6_Ccu_Isr
Syntax:	void Icu_17_GtmCcu6_Ccu_Isr (uint8 CcuKernalNumber,)
Service ID:	0x1A
Sync/ Async:	Synchronous

Origin	Vendor specific	
Re-entrancy:	Reentrant	
Parameters (in):	CcuKernalNumber	Ccu6 Module number
Parameters (out):	None	
Return value:	None	
Description:	ISR to handle CCU requests. CCU resources can handle edge detection and signal measurement mode. Report safety error ICU_E_INTR_SRC_ERROR in case of interrupt source is wrong.	
Caveats:	None	
Configuration:	None	
DET:	None	
DEM:	None	
Implementation Comments:	None	
I/O Mode	This API can be invoked when the CPU is in either User-1 Mode or Supervisor Mode.	

6.5 Exclusive Area (EA) Information

The exclusive areas (critical sections) are provided for any of the following reason:

- 1) To protect shared resources from the module interrupts. The resources are used in ISR's.
- 2) To protect shared resources from another API / same API.

Note: All exclusive areas implemented in ICU are to protect shared resources from the module interrupts.

6.5.1 ICU_EA_RESET_EDGECOUNT

Table 118 Specification of EA ICU_EA_RESET_EDGECOUNT

API Name:	Icu_17_GtmCcu6_ResetEdgeCount
Resource Name:	• "EdgeCountValuePtr" structure member of Icu_17_GtmCcu6_AsilGtmVarlist and Icu_17_GtmCcu6_QmVarlist. • TIM registers
ISR Name:	NA
Comment:	This exclusive area invariably protects TIM registers as well

6.5.2 ICU_EA_INT_NOTIFY

Table 119 Specification of EA ICU_EA_INT_NOTIFY

API Name:	Icu_17_GtmCcu6_EnableWakeup, Icu_17_GtmCcu6_EnableNotification
Resource Name:	• "GlbIndexStatus" structure member of Icu_17_GtmCcu6_AsilGtmVarlist, Icu_17_GtmCcu6_AsilCcu6Varlist and Icu_17_GtmCcu6_QmVarlist. • TIM register
ISR Name:	NA

Comment:	This exclusive area assures consistent updated of bothTIM register and status in "GlbIndexStatus". Losing this synchronization can cause indeterministic behavior.
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6.5.3 ICU_EA_ENABLEWAKEUP

Table 120 Specification of ICU_EA_ENABLEWAKEUP

API Name:	Icu_17_GtmCcu6_EnableWakeup
Resource Name:	"GlbIndexStatus" structure member of Icu_17_GtmCcu6_AsilGtmVarlist, Icu_17_GtmCcu6_AsilCcu6Varlist and Icu_17_GtmCcu6_QmVarlist. - TIM register
ISR Name:	NA
Comment:	This exclusive area assures consistent updation of bothTIM register and status in Icu_GlbIndexStatus. Losing this synchronization can cause indeterministic behavior. .

6.5.4 ICU_EA_CCU6_IENUPDATE

Table 121 Specification of ICU_EA_CCU6_IENUPDATE

API Name:	Icu_17_GtmCcu6_SetActivationCondition
Resource Name:	IEN register
ISR Name:	NA
Comment:	This exclusive area assures consistent updation of IEN register of CCU6x . The local function Icu_ICcu6SetActivationCondition() called from Icu_17_GtmCcu6_SetActivationCondition() which does the updation of CCU6x IEN register.

6.5.5 ICU_EA_CCU6_INTHANDLE

Table 122 Specification of ICU_EA_CCU6_INTHANDLE

API Name:	NA
Resource Name:	IS and ISR register of CCU6x
ISR Name:	Icu_17_GtmCcu6_Ccu_Isr
Comment:	This exclusive area assures consistent updation of ISR register of CCU6x .

6.5.6 ICU_EA_CCU6_VARUPDATE

Table 123 Specification of ICU_EA_CCU6_VARUPDATE

API Name:	Icu_17_GtmCcu6_GetTimeElapsed
Resource Name:	"CcuRRegVal" structure member of Icu_17_GtmCcu6_QmVarlist and Icu_17_GtmCcu6_AsilCcu6Varlist. - SR and R value of CCU6x.
ISR Name:	NA
Comment:	This exclusive area assures consistent updation of "CcuRRegVal" structure member of Icu_17_GtmCcu6_QmVarlist and Icu_17_GtmCcu6_AsilCcu6Varlist. The local function Icu_ICcu6GetTimeElapsed () called from Icu_17_GtmCcu6_GetTimeElapsed() for the updation.

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6.5.7 ResetEdgeCount

Table 124 Specification of ResetEdgeCount

API Name:	Icu_17_GtmCcu6_ResetEdgeCount
Resource Name:	“EdgeCountValuePtr” structure member of Icu_17_GtmCcu6_AsilGtmVarlist and Icu_17_GtmCcu6_QmVarlist. - TIM register
ISR Name:	NA
Comment:	This exclusive area invariably protects TIM register as well.

6.5.8 EnableWakeup

Table 125 Specification of EnableWakeup

API Name:	Icu_17_GtmCcu6_EnableWakeup
Resource Name:	“GlbIndexStatus” structure member of Icu_17_GtmCcu6_AsilGtmVarlist, Icu_17_GtmCcu6_AsilCcu6Varlist and Icu_17_GtmCcu6_QmVarlist. - TIM register
ISR Name:	NA
Comment:	This exclusive area assures consistent updation of bothTIM register and status in GlbIndexStatus. Losing this synchronization can cause indeterministic behavior. .

6.5.9 EnableNotification

Table 126 Specification of EnableNotification

API Name:	Icu_17_GtmCcu6_EnableNotification
Resource Name:	“GlbIndexStatus” structure member of Icu_17_GtmCcu6_AsilGtmVarlist, Icu_17_GtmCcu6_AsilCcu6Varlist and Icu_17_GtmCcu6_QmVarlist. - TIM register
ISR Name:	NA
Comment:	This exclusive area assures consistent updation of bothTIM register and status in GlbIndexStatus. Losing this synchronization can cause indeterministic behavior. .

6.5.10 Ccu6IenUpdate

Table 127 Specification of Ccu6IenUpdate

API Name:	Icu_17_GtmCcu6_SetActivationCondition
Resource Name:	IEN register
ISR Name:	NA
Comment:	This exclusive area assures consistent updation of IEN register of CCU6x . The local function Icu_ICcu6SetActivationCondition() called from Icu_17_GtmCcu6_SetActivationCondition() which does the updation of CCU6x IEN register.

6.5.11 CcuInterruptHandle

Table 128 Specification of CcuInterruptHandle

API Name:	NA
Resource Name:	IS and ISR register of CCU6x
ISR Name:	Icu_17_GtmCcu6_Ccu_Isr
Comment:	This exclusive area assures consistent updation of ISR register of CCU6x .

6.5.12 CcuVariableupdate

Table 129 Specification of CcuVariableupdate

API Name:	Icu_17_GtmCcu6_GetTimeElapsed
Resource Name:	• “CcuRRegVal” structure member of Icu_17_GtmCcu6_QmVarlist and Icu_17_GtmCcu6_AsilCcu6Varlist. • SR and R value of CCU6x.
ISR Name:	NA
Comment:	This exclusive area assures consistent updation of “CcuRRegVal” structure member of Icu_17_GtmCcu6_QmVarlist and Icu_17_GtmCcu6_AsilCcu6Varlist. The local function Icu_ICcuGetTimeElapsed () called from Icu_17_GtmCcu6_GetTimeElapsed() for the updation.

7 Error Classification

Table 130 Error Classification

Error Code	Error	Relevance	Value (hex)	Used for DET check	Used for safety check
ICU_E_PARAM_CONFIG	The Parameter passed is not within range (Not a NULL Pointer)	Development and Safety	0x0A	Yes	Yes (validating the configuration marker)
ICU_E_PARAM_CHANNEL	Invalid Channel Number Or the channel is not configured for the required measurement mode.	Development and Safety	0x0B	Yes	Yes
ICU_E_PARAM_ACTIVATION	Invalid Activation parameter in API Icu_17_GtmCcu6_SetActivationCondition	Development and Safety	0x0C	Yes	Yes
ICU_E_PARAM_BUFFER_POINTER	Invalid Buffer Pointer for APIs Icu_17_GtmCcu6_StartTimeStamp and Icu_17_GtmCcu6_GetDutyCycleValues	Development and Safety	0x0D	Yes	Yes
ICU_E_PARAM_BUFFER_SIZE	Invalid Buffer Size used in API	Development	0x0E	Yes	Yes

Error Code	Error	Relevance	Value (hex)	Used for DET check	Used for safety check
	Icu_17_GtmCcu6_StartTime stamp	and Safety			
ICU_E_PARAM_MODE	Invalid Mode is passed for the API Icu_17_GtmCcu6_SetMode	Development and Safety	0x0F	Yes	Yes
ICU_E_UNINIT	API service used without module initialization	Development	0x14	Yes	No
ICU_E_NOT_STARTED	API Service Icu_17_GtmCcu6_StopTime stamp called before starting the measurement	Development	0x15	Yes	No
ICU_E_BUSY_OPERATION	Icu_17_GtmCcu6_SetMode is called when a channel is in running condition	Development	0x16	Yes	Yes
ICU_E_ALREADY_INITIALIZED	API Icu_17_GtmCcu6_Init service called when the ICU driver and the Hardware are already initialized	Development	0x17	Yes	No
ICU_E_PARAM_VINFO	Icu_17_GtmCcu6_GetVersionInfo called with a parameter NULL_PTR	Development	0x19	Yes	No
ICU_E_PARAM_NOTIFY_INTERVAL	Icu_17_GtmCcu6_StartTime Stamp called with parameter NotifyInterval is invalid (e.g."0", NotifyInterval < 1)	Development	0x18	Yes	Yes
ICU_E_NO_WAKEUP	Safety specific error value reported when 'Icu_SetMode' called to SLEEP without IRQ enabled for the wakeup channel which is Started and wakeup enabled	Safety relevant	0x20	No	Yes
ICU_E_PARAM_CLC_ERROR	The CCU6 clock initialization is failed in function Icu_17_GtmCcu6_Init	Safety relevant	0x35	No	Yes
ICU_E_INTR_SRC_ERROR	Error reporting in case of unexpected triggering of service requests or call of a wrong ISR	Safety relevant	0x36	No	Yes

8 Example Usage

This chapter describes how the ICU Driver can be configured and how to use different API provided by it.

8.1 Configuration of the Driver

To configure ICU Driver the following guidelines should be followed properly.

1. MCU Driver: Configure the System Clock. In ERU, configure EruChannelUsage to "USED_BY_ICU_DRIVER" & select EruInputLine the Input ports for the used ERU channels.

2. GTM Driver: Configure GTM Clocks(CMU). Configure Clock and Port selection for the used TIM channels. Configure TBU channel which will be needed for Channels in Timestamp measurement mode.
3. PORT Driver: For all the port pins that were used in GTM Driver as input pins, configure the same in PORT Driver. Set the direction to input and port function as GTM.
4. Interrupt Priority Initialization: The priority of interrupts should be configured in IRQ Driver and appropriate priorities should be given.
5. ICU Driver: Select the required API configuration and choose channel dependent parameters like IcuDefaultStartEdge, IcuMeasurementMode etc.

8.1.1 Sample configuration of the Driver

Requirement:

The user requires 20 QM channels to be configured in ICU Driver. 9 channels for signal measurement, 5 channels for edge detection, 4 channels for timestamp capture and 2 channels for edge counter. The user wishes to use all API's but for Icu_17_GtmCcu6_Delnit API. The following table shows the channel number and allotted resources

Table 131 Channel configuration

Channel Number	IcuAssignedHwUnit	IcuMeasurementMode
0	ERU6	ICU_MODE_SIGNAL_EDGE_DETECT ICU_BOTH_EDGES
1	TIM0 Channel 1	ICU_MODE_SIGNAL_EDGE_DETECT ICU_BOTH_EDGES
2	TIM3 Channel 7	ICU_MODE_SIGNAL_EDGE_DETECT ICU_RISING_EDGE
3	TIM3 Channel4	ICU_MODE_SIGNAL_MEASUREMENT ICU_FALLING_EDGE IcuSignalMeasurement, <START AS40> IcuActiveTime<END AS40><START AS321>IcuDutyCycle<END AS321>
4	TIM3 Channel3	ICU_MODE_TIMESTAMP ICU_BOTH_EDGES
5	TIM2 Channel5	ICU_MODE_SIGNAL_MEASUREMENT ICU_RISING_EDGE
6	ERU7	ICU_MODE_SIGNAL_EDGE_DETECT ICU_BOTH_EDGES
7	ERU0	ICU_MODE_SIGNAL_EDGE_DETECT ICU_BOTH_EDGES
8	TIM2 Channel0	ICU_MODE_TIMESTAMP ICU_BOTH_EDGES
9	TIM1 Channel4	ICU_MODE_TIMESTAMP ICU_BOTH_EDGES
10	TIM0 Channel7	ICU_MODE_TIMESTAMP ICU_BOTH_EDGES
11	TIM3 Channel6	ICU_MODE_EDGE_COUNTER
12	TIM3 Channel0	ICU_MODE_SIGNAL_MEASUREMENT

Channel Number	IcuAssignedHwUnit	IcuMeasurementMode
		ICU_FALLING_EDGE
13	TIM2 Channel7	ICU_MODE_SIGNAL_MEASUREMENT ICU_FALLING_EDGE
14	TIM1 Channel5	ICU_MODE_SIGNAL_MEASUREMENT ICU_RISING_EDGE
15	Ccu60/T12/Cc60	ICU_MODE_SIGNAL_MEASUREMENT ICU_RISING_EDGE
16	Ccu60/T12/Cc61	ICU_MODE_SIGNAL_MEASUREMENT ICU_FALLING_EDGE
17	Ccu60/T12/Cc62	ICU_MODE_SIGNAL_MEASUREMENT ICU_FALLING_EDGE
18	Ccu61/T12/Cc60	ICU_MODE_SIGNAL_MEASUREMENT ICU_FALLING_EDGE
19	Ccu61/T12/Cc61	ICU_MODE_SIGNAL_EDGE_DETECT ICU_FALLING_EDGE

IcuCodeOptimizeSwitches:

As per the above configuration the following should be done in configuration header file ([Code Listing 1](#)) to reduce the ROM space. The following should be carefully observed

TIM is used in edge detection, edge counter, signal measurement and time stamp capture modes.

Code Listing 1 Optimize Switches for configuration

```
001: #define ICU_TIM_SIGMEAS_USED      (ON)
002: #define ICU_TIM_SIGEDGE_USED      (ON)
003: #define ICU_TIM_TIMSTMP_USED      (ON)
004: #define ICU_TIM_EDGECONT_USED      (ON)
005: #define ICU_ERU_SIGEDGE_USED      (ON)
```

Sample configuration of Gtm Tim Channel:

TimChannelGeneral:

```
TimChannelUsage: USED_BY_OTHER_DRIVERS_OR_UNUSED
TimChannelClockSelect: GTM_CONFIGURABLE_CLOCK_2
TimChannelGpr0InputSelect: TIMEBASE_TBU_TS0
TimChannelTbu0Ts0BitsSelect: BITS_0_TO_23
```

TimChannelFilterConfig:

```
TimChFilterCounterFreqSelect: GTM_CONFIGURABLE_CLOCK_0
TimChFilterModeForRisingEdge: IMMEDIATE_EDGE_PROPAGATION_MODE
TimChFilterModeForFallingEdge: IMMEDIATE_EDGE_PROPAGATION_MODE
TimChannelFilterEnable : Enable or disable
TimChFilterTimeForRisingEdge
TimChFilterTimeForFallingEdge
```

TimChannelInterrupt:

TimNotification: Icu_17_GtmCcu6_Isr

Port pin configuration:

TIM0 Channel0 to Channel1 → Input, select available port pin

For all these pins under port configuration select the pin direction as Input and if required choose pull up or pull down as per configuration.

Icu Channel Configuration: The following [Code Listing 2](#) shows sample configuration for channel 0.

Code Listing 2 Channel 1 configuration

```
{
    /* ICU Channel 1 */
    &Icu_Notification1, /*Notification-function name*/
    {
        /*Measurement Mode*/
        ICU_MODE_SIGNAL_EDGE_DETECT,

        /*Default Start Edge */
        ICU_BOTH_EDGES,

        /*Measurement Property*/
        0U,

        /*Wakeup capability*/
        ICU_NOT_WAKEUPCAPABLE,

        /* Assigned Hardware Resource Number */
        GTM_TIM0_CHANNEL1,

        /* Assigned Hw Unit */
        (uint8)ICU_GTM_OPTION,

        /* CCU connection */
        0U,

        /* Safety Signal or QM signal */
        ICU_QM_SIGNAL,

        0U, /* Reserved */

#ifdef (ICU_SAFETY_ENABLE == STD_ON)
        0U, /*Buffer Marker value */
#endif

        0U, /* Filtering time for rising edge */

        0U, /* Filtering time for falling edge */

        GTM_INTERRUPT_LEVEL_MODE, /* Filtering time for TIM interrupt mode */

        GTM_CONFIGURABLE_CLOCK_2, /*Channel Clock Select */

        TIMEBASE_TBU_TS0, /* Gpr0 input select for the Tim channel */

        1U, /* Enables Filter for the Channel */

        /*Decides the filter Counter frequency for the TIM Channel */
        GTM_CONFIGURABLE_CLOCK_0,
```


Code Listing 2 Channel 1 configuration

```

/*Decides the filter mode for rising Edge of the TIM Channel Input */
IMMEDIATE_EDGE_PROPAGATION_MODE,
/*Decides the filter mode for falling Edge of the TIM Channel Input */
IMMEDIATE_EDGE_PROPAGATION_MODE,
}
},

```

The Icu_17_GtmCcu6_ConfigType should also be filled appropriately, there are 2 channels in each measurement mode and ERU interrupt sources are not used. This should be notified to the driver accordingly.

Code Listing 3 Icu_17_GtmCcu6_ConfigType configuration

```

const Icu_17_GtmCcu6_ConfigType Icu_ConfigRoot[] =
{
{
/*Pointer to channel configuration set */
&Icu_kChannelConfig0[0],

/* Pointer to Phsyscial channel mapping set */
&Icu_17_GtmCcu6_MappingIndex0[0],

#if (ICU_REPORT_WAKEUP_SOURCE == STD_ON)
&Icu_kChannelWakeup0[0],
#endif
#if (ICU_ERU_SIGEDGE_USED == STD_ON)
{
7U,
ERU_REQLINE_UNUSED,
ERU_REQLINE_UNUSED,
ERU_REQLINE_UNUSED,
ERU_REQLINE_UNUSED,
ERU_REQLINE_UNUSED,
0U,
6U
},
#endif
#if (ICU_CCU6_USED == STD_ON)
/* T12ClkSelection -> Lower 3 bits (Bits 0 to 2)
T12PrescalerEnabled -> Bit 3
Ccu6SleepEnable -> Bit 4
Initialized by ICU or not -> Bit 7. ICU Initialization -> 0,
Other Driver Initialization -> 1
*/
{
0U,
0U,
},
#endif
/*
Enter the no of channels configured in the following order
1. Number of Signal Edge Detection Channels.
2. Number of Timestamp channels
*/

```

Code Listing 3 Icu_17_GtmCcu6_ConfigType configuration

```

    3. Number of EdgeCount Channels
    4. Number of Signal Measurement Channels.
    */
    {5U, 4U, 2U, 9U}

}
};

```

Wakeup channel configuration is given below

The configured channel are divided into three groups ASIL-GTM, ASIL-CCU6 and QM. The MappingIndex maps the configured channel order to either of the three sets ASIL-GTM, ASIL-CCU6 or QM channels(GTM/CCU6/ERU)

Code Listing 4 Icu_17_GtmCcu6_MappingIndex generated out of Configured channels

```

const Icu_17_GtmCcu6_MapType Icu_17_GtmCcu6_MappingIndex0[ICU_MAX_CHANNELS] =
{
    /*IcuChannel_0*/
    {
        0U,
        /* Wakeup capable, QM Signal */
        0U
    },
    /*IcuChannel_1*/
    {
        1U,
        /* Edge Detection, Indexing not required */
        0U
    },
    /*IcuChannel_2*/
    {
        2U,
        /* Edge Detection, Indexing not required */
        0U
    },
    /*IcuChannel_3*/
    {
        3U,
        /* Signal Measurement, QM Signal */
        0U
    },
    /*IcuChannel_4*/
    {
        4U,
        /* Time Stamp, QM Signal */
        0U
    },
    /*IcuChannel_5*/
    {
        5U,
        /* Signal Measurement, QM Signal */
        1U
    },
    /*IcuChannel_6*/
    {
        6U,

```

Code Listing 4 Icu_17_GtmCcu6_MappingIndex generated out of Configured channels

```

/* Edge Detection, Indexing not required */
0U
},
/*IcuChannel_7*/
{
    7U,
    /* Edge Detection, Indexing not required */
    0U
},
/*IcuChannel_8*/
{
    8U,
    /* Time Stamp, QM Signal */
    1U
},
/*IcuChannel_9*/
{
    9U,
    /* Time Stamp, QM Signal */
    2U
},
/*IcuChannel_10*/
{
    10U,
    /* Time Stamp, QM Signal */
    3U
},
/*IcuChannel_11*/
{
    11U,
    /* Edge Count, QM Signal */
    1U
},
/*IcuChannel_12*/
{
    12U,
    /* Signal Measurement, QM Signal */
    2U
},
/*IcuChannel_13*/
{
    13U,
    /* Signal Measurement, QM Signal */
    3U
},
/*IcuChannel_14*/
{
    14U,
    /* Signal Measurement, QM Signal */
    4U
},
/*IcuChannel_15*/
{
    15U,
    /* Signal Measurement, QM Signal */
    5U
},
/*IcuChannel_16*/

```

Code Listing 4 Icu_17_GtmCcu6_MappingIndex generated out of Configured channels

```
{
    16U,
    /* Signal Measurement, QM Signal */
    6U
},
/*IcuChannel_17*/
{
    17U,
    /* Signal Measurement, QM Signal */
    7U
},
/*IcuChannel_18*/
{
    18U,
    /* Signal Measurement, QM Signal */
    8U
},
/*IcuChannel_19*/
{
    19U,
    /* Edge Detection, Indexing not required */
    0U
}
};
```

Wakeup channel configuration is given below

Code Listing 5 Wakeup configuration

```
const Icu_ChannelWakeupType Icu_kChannelWakeup0 [] =
{
    {
        {123U}, /*Wakeup Reason Value for Channel 1*/
    },
};
```

8.2 Driver Initialization

The [Code Listing 6](#) depicts the steps involved in the initialization of the ICU Driver. Note: User must take care that the Icu_17_GtmCcu6_Init API is called before using any other API provided by ICU Driver. Mcu_Init must be called before Icu_17_GtmCcu6_Init, it will initialize the GTM resources. Similarly call IrqGtm_init, IrqCcu6_Init and IrqEru_Init functions for initializing the interrupt priorities. The user need not call these functions explicitly if Ecum Driver is configured to have the modules mentioned.

Code Listing 6 Driver Initialization

```
006:      Mcu_Init(&Mcu_ConfigRoot[0]);
007:      Mcu_InitClock(0);
008:      while(Mcu_GetPllStatus(0) == 0);
009:      Mcu_DistributePllClock();
010:      /* Configure Interrupt priority */
011:      IrqGtm_Init();
012:      /* Configure interrupt priority */
013:      IrqCcu6_Init();
014:      /* Configure Interrupt priority */
015:      IrqScu_Init();
```

Code Listing 6 Driver Initialization

```
016:      /* Port Initialization */
017:      Port_Init(&Port_ConfigRoot[0]);
018:      /* Icu Initialization */
019:      Icu_17_GtmCcu6_Init(&Icu_ConfigRoot[0]);
020:      /* Enable All Interrupts */
021:      EnableAllInterrupts();
022:      /* Continue with the Driver */
023:
```

8.3 API Usages

8.3.1 Edge detection and notification

The following code snippet shows enabling / disabling notification. As per the provided example channel 6 and channel 7 are used for signal edge detection / notification.

Code Listing 7 Edge detection

```
024:      Icu_17_GtmCcu6_SetActivationCondition(6, ICU_RISING_EDGE);
025:      Icu_17_GtmCcu6_EnableNotification(1);
026:      Icu_17_GtmCcu6_DisableNotification(1);
027:      ...
```

To disable / re enable the already running edge detection channel.

Code Listing 8 Notification

```
028:      Icu_17_GtmCcu6_DisableEdgeDetection (1);
029:      Icu_17_GtmCcu6_EnableEdgeDetection (1);
030:      ...
```

8.3.2 EnableMultiEdgeDetection

The following code snippet shows the generate interrupt in ICU edge detection measurement mode after specified number of edges using `Icu_17_GtmCcu6_EnableMultiEdgeDetection()` API:
Channel 2 must be configured as edge detection mode.

Code Listing 9 Multi edge detection

```
031:      Icu_17_GtmCcu6_EnableMultiEdgeDetection(2,2);
032:      Icu_17_GtmCcu6_EnableNotification(2);
033:
034:      /* To disable the multi edge detection mode */
035:      Icu_17_GtmCcu6_DisableEdgeDetection(2);
```

8.3.3 Timestamp

The following code snippet shows how to use Timestamp APIs.

Code Listing 10 Driver Set Period and Duty Cycle

```
036:      /* enable notification for a timestamp channel */
037:      Icu_17_GtmCcu6_EnableNotification(4);
038:      /* start time stamp capture , give notification after 10 captures */
```

Code Listing 10 Driver Set Period and Duty Cycle

```
039: Icu_17_GtmCcu6_StartTimestamp(4,Buffer,100,10);
040: ...
041: Icu_17_GtmCcu6_StopTimestamp(4);
042:
```

8.3.4 Edge Counter

The following [Code Listing 11](#) shows how to use edge counter APIs.

Code Listing 11 Edge Counter API's

```
043: /* Enable edge count for channel 11 */
044: Icu_17_GtmCcu6_EnableEdgeCount(11);
045: /* Get how many edges are there*/
046: Edges = Icu_17_GtmCcu6_GetEdgeNumbers(11);
047: /* Disable Edge count */
048: Icu_17_GtmCcu6_DisableEdgeCount(11);
049: /* Put the edge counter to 0 */
050: Icu_17_GtmCcu6_ResetEdgeCount(11);
051:
```

8.3.5 Signal Measurement

The following [Code Listing 12](#) shows how to read measured signal values. Signal measurement starts after calling Icu_17_GtmCcu6_StartSignalMeasurement.

Code Listing 12 Signal measurement API's

```
052: Icu_17_GtmCcu6_ValueType Time;
053: Icu_17_GtmCcu6_InputStateType State;
054: Icu_17_GtmCcu6_DutyCycleType Duty;
055: Icu_17_GtmCcu6_StartSignalMeasurement(3);
056: State = Icu_17_GtmCcu6_GetInputState(3);
057: If (State == ICU_ACTIVE)
058: {
059:     Time = Icu_17_GtmCcu6_GetTimeElapsed(3);
060: }
061: State = Icu_17_GtmCcu6_GetInputState(3);
062: If (State == ICU_ACTIVE)
063: {
064:     Icu_17_GtmCcu6_GetDutyCycleValues(3,&Duty);
065: }
066:
067:
```

8.3.6 Set Mode and wakeup

There is one wakeup channel configured in our sample configuration (refer [Code Listing 5](#)). The following code snippet illustrates how to use the wakeup feature.

Code Listing 13 Set Mode and Wakeup

```
001: Icu_17_GtmCcu6_Init(&Icu_ConfigRoot[0]);
```

Code Listing 13 Set Mode and Wakeup

```

002:  /* After Icu_17_GtmCcu6_Init the all wakeup events are disabled */
003:  Icu_17_GtmCcu6_EnableWakeup(ICU_CHANNEL_1); /* channel 1 */
004:  /* Channel 6 is made Wakeup capable */
005:  Icu_17_GtmCcu6_SetMode(ICU_MODE_SLEEP) ;
006:  /* Set the mode to sleep */
007:  ..
008:  ..
009:  Mcu_SetMode(MCU_SLEEP)
010:  /* process in sleep */
011:  /* Generate trigger from port pin configured for TIM0 Channel1 , alerts
    TIM0 Channel1, the interrupt in TIM0 Channel1 executes */
012:  /* The control from the ICU ISR will be received in EcuM_CheckWakeup
    (123) parameter 123 is the configured wakeup info. */

```

9 Limitations and Assumptions

9.1 Assumptions and Deviations from Software Specification

The following software assumptions are made in implementing ICU Driver.

Table 132 Assumptions

Requirement ID	APIs affected	Description
None	Icu_17_GtmCcu6_GetInputState/ Icu_17_GtmCcu6_GetTimeElapsed/ Icu_17_GtmCcu6_GetDutyCycleValues	Availability of a new data is not only applicable from the instant that particular API was invoked prior or from the time of Icu_17_GtmCcu6_Init, but also from the time, the channel is enabled in the corresponding measurement mode. i.e. by a call of Icu_17_GtmCcu6_EnableEdgeDetection or Icu_17_GtmCcu6_StartSignalMeasurement.
ICU295 of AS4.0.3	Icu_17_GtmCcu6_GetTimeElapsed, Icu_17_GtmCcu6_GetDutyCycleValues	ICU_DUTY_CYCLE and its related functionalities are not provided. A discussion is ongoing on this in Bugzilla 52843
ICU251 and ICU252 (of AS4.0.3)	Icu_17_GtmCcu6_EnableNotifications, Icu_17_GtmCcu6_DisableNotification	These 2 requirements are not implemented. Please refer Bugzilla 52388 for details.

The Following Bugzilla are considered in implementing ICU Driver for AS3.2.1

- 40686: IcuWakeupFunctionalityApi is considered for AS3.2.1
- 46551: IcuEdgeDetectApi & its functionality considered for AS3.2.1
- 29836: NULL Pointer check in Icu_17_GtmCcu6_GetVersionInfo is considered for AS3.2.1
- 29830: Req ICU313 "Icu_GetInputState shall return Icu_17_GtmCcu6_InputStateType which will have value ICU_IDLE,
when no activation edge has been detected since the last call of Icu_GetInputState () or Icu_Init ()"

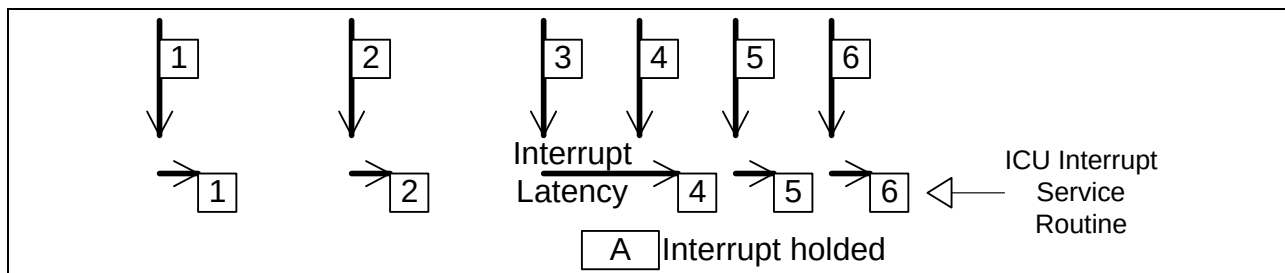
Of AS402 is considered. Discarded ICU033 of AS3.2.1.

5. 14377: Edge detection functionality shall be pre compile time configurable On/Off
6. 41025: Multiplicities of parameters "IcuSignalNotification" & "IcuTimestampNotification" are changed to 0...1.
7. 29823: Two Paragraphs in Sec 8.2.4 are ignored.
8. 30210, 40683 & 25392: Considered ICU213 & ICU228 of AS402. Req of AS321 ignored.
9. 41104: Removed container IcuEdgeCounterMeasurement.
10. 42968: Ranges of parameters changed.
 - IcuMaxChannel – 0 to 65535
 - IcuChannelId - 0 to 65535
 - IcuIndex – 0 to 255.
11. 41553: Multiplicity of IcuChannelWakeupInfo changed to 0...1

9.2 Limitations

During Edge detection, if edges occurred on the same channel faster than the Interrupt latency, driver will not provide missed edge notifications.

Figure 7 Software limitation in detecting edges



In the above **Figure 7**, it can be observed that, when edge 3 occurs, because of interrupt latency and before executing ICU interrupt routine, the edge 4 occurs. In this scenario user will get notification only one time. The above logic also applies for signal measurement, timestamp measurement and edge count Measurement, if the input frequency is very high (higher than the run time of ISR); there is a possibility of obtaining a wrong result.

During signal measurement of ICU_HIGH_TIME, ICU_LOW_TIME and ICU_ACTIVE_TIME/ICU_DUTY_CYCLE, if there is a delayed interrupt there is a possibility of getting a wrong result and the behavior will continue until reset.

Software configuration limitations:

- Please note, OGU0 of ERU is used for ICU purpose and hence no other module should use this resource, if any channel is configured using ERU resource.
- The port pin allocation should be done in such a way that it does not conflict with other allocations.
- For each Icu Channel Config Set, there has to be one corresponding Gtm Configuration/Mcu Module Config Set. So that any channel of an ICU Config set shall use Gtm Tim or ErU Channels belongs to its corresponding Mcu Config Set only.
- Resource clash within a config set is handled by configuration tool. An error is asserted if the same TIM or ERU channel is being used by different ICU channels within a config set.
- For resource conflict avoidance in usage of CCU6 CC6x channels, one whole kernel (i.e. 3 channels) will be allocated for a driver. This would mean, if CCU60 CC61 is planned to be used for ICU purpose, then CCU60 as a whole has to be allocated to ICU. In such a case, it is ICU's responsibility to configure the T12 timers and the configurations of the channels present in that. The other channels in that kernel should not be used by any other driver.

- In case of measurement using CCU6, only 1 T12 timer overflow is handled. Multiple overflows will result in wrong value. Also, in case of overflow, the value should be within the next time the value FirstVal is reached. It has to be ensured by the user to check that the input signal does not span more than 1 complete T12 timer range.

9.3 Hardware Features not supported

The hardware GTM TOM module output can be routed to TIM module. Currently this feature is not implemented in ICU driver.

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